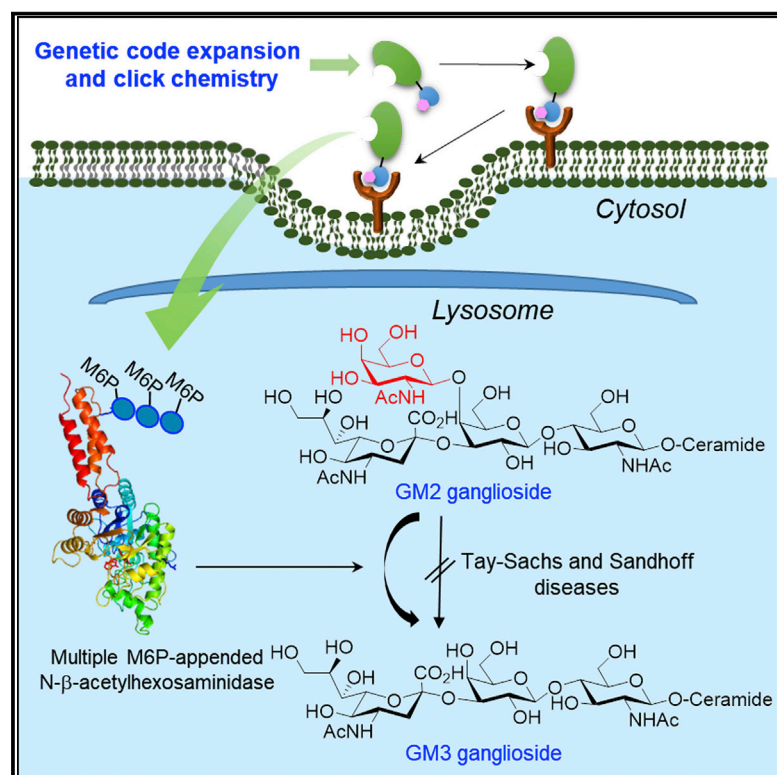


Cell Chemical Biology

A Glycoengineered Enzyme with Multiple Mannose-6-Phosphates Is Internalized into Diseased Cells to Restore Its Activity in Lysosomes

Graphical Abstract



Authors

Ji Young Hyun, Sanggil Kim,
Hyun Soo Lee, Injae Shin

Correspondence

hslee76@sogang.ac.kr (H.S.L.),
injae@yonsei.ac.kr (I.S.)

In Brief

A glycoengineered lysosomal enzyme containing multiple mannose-6-phosphates was prepared by combining genetic code expansion with bioorthogonal ligation techniques. The glycoengineered enzyme was efficiently internalized into diseased cells and accumulated into lysosomes. In addition, the enzyme successfully removed GM2-ganglioside stored in lysosomes of diseased cells.

Highlights

- The multi-M6P-containing lysosomal enzyme is prepared by using two techniques
- A glycoengineered enzyme is efficiently internalized into diseased cells
- A glycoengineered enzyme removes GM2-ganglioside stored in lysosomes of cells
- A lysosome-targeting fluorogenic substrate is prepared to monitor enzyme activity



A Glycoengineered Enzyme with Multiple Mannose-6-Phosphates Is Internalized into Diseased Cells to Restore Its Activity in Lysosomes

Ji Young Hyun,¹ Sanggil Kim,² Hyun Soo Lee,^{2,*} and Injae Shin^{1,3,*}

¹Center for Biofunctional Molecules, Department of Chemistry, Yonsei University, Seoul 03722, Republic of Korea

²Department of Chemistry, Sogang University, Seoul 04107, Republic of Korea

³Lead Contact

*Correspondence: hlee76@sogang.ac.kr (H.S.L.), injae@yonsei.ac.kr (I.S.)

<https://doi.org/10.1016/j.chembiol.2018.07.011>

SUMMARY

In this study we developed an efficient method to prepare glycoengineered β -N-acetylhexosaminidase containing multiple mannose-6-phosphates (M6Ps) by combining genetic code expansion with bioorthogonal ligation techniques. We found that multiple M6P-conjugated enzymes were produced with a high efficiency by using combined techniques. Importantly, glycoengineered enzymes entered lysosomes of patient-derived primary cells, which lack endogenous lysosomal β -N-acetylhexosaminidase, more readily than commercialized human β -hexosaminidase. Moreover, glycoengineered enzymes successfully removed GM2-ganglioside stored in lysosomes of diseased cells, indicating that its activity is restored in diseased cells. We also synthesized and applied a lysosome-targeting fluorogenic substrate to monitor endogenous and supplemental glycoengineered β -N-acetylhexosaminidase activities in lysosomes. The results of this study indicate that the present strategy, which relies on genetic code expansion and bioorthogonal ligation techniques, is highly attractive to generate multi-M6P-containing lysosomal enzymes that can be used to study lysosomal storage disorders associated with lysosomal enzyme deficiencies.

INTRODUCTION

Lysosomal acid hydrolases are crucial for degradation and recycling of various biomolecules, including (glyco)proteins, glycosphingolipids, and nucleic acids (Luzio et al., 2007). These enzymes are produced in the form of glycoproteins conjugated with oligosaccharides containing terminal mannose-6-phosphate (M6P) (Kornfeld, 1987; Braulke and Bonifacino, 2009). The hydrolases are then translocated to lysosomes by initially entering the prelysosomal compartment, called the endosome, through the interaction with the membrane-bound M6P receptor (M6PR) and subsequent fusion of the endosome with the lysosome. However, some of the enzymes evade lysosomal traf-

ficking and are subjected to the secretory pathway for secretion from the cell. The secreted enzymes can be internalized into the cell by binding to the cell-surface M6PR via receptor-mediated endocytosis and again enter the lysosome (Braulke and Bonifacino, 2009; Ghosh et al., 2003).

The two M6PRs, a smaller cation-dependent M6PR (CD-M6PR) and a larger cation-independent M6PR (CI-M6PR, also known as an insulin-like growth factor-II [IGF-II] receptor), have been identified (Ghosh et al., 2003; Dahms et al., 2008). They play a critical role in transporting synthesized lysosomal enzymes from the Golgi apparatus to lysosomes by binding to M6P moieties linked to the enzymes. The CI-M6PR is also present on the cell surface and acts as a receptor to re-capture lysosomal enzymes that are secreted from the cell. Thus, the CI-M6PR is a major receptor that is associated with the cellular uptake of extracellular lysosomal enzymes appended by M6P. The extracytoplasmic domain of CI-M6PR possesses two M6P-binding sites (also a single IGF-II-binding site), and the CD-M6PR contains a single M6P-binding site. It is known that while the two M6PRs bind to monomeric M6P with similar affinities, the CI-M6PR interacts with the high-mannose oligosaccharide containing two phosphate groups more strongly than the CD-M6PR (Tong and Kornfeld, 1989). This indicates that the CI-M6PR binds with more avidity to multimeric M6Ps than does the CD-M6PR.

Genetic deficiencies of lysosomal enzymes lead to the inability of the lysosome to digest biomolecular substances. The deficiencies result in the progressive accumulation of undigested substances in lysosomes and promote the onset of lysosomal storage disorders (LSDs) (Futerman and van Meer, 2004). For example, defects in the digestion of glycosphingolipids cause GM1-gangliosidosis, GM2-gangliosidosis (Tay-Sachs and Sandhoff diseases), Gaucher and Fabry diseases, and other maladies (Jeyakumar et al., 2002; Wennekes et al., 2009; Platt, 2014). The clinical method for treatment of LSDs includes exogenous supplementation of active enzymes, known as enzyme replacement therapy (ERT) (Solomon and Muro, 2017). In this therapy, M6P-oligosaccharide-containing enzymes are needed to efficiently enter cells via M6PR-mediated endocytosis and to accumulate in lysosomes.

The active enzymes used in ERT are typically obtained from mammalian cells (Solomon and Muro, 2017; Crawley et al., 1997; He et al., 2012). However, the enzymes produced in cells contain relatively low levels of M6P and heterogeneous glycans



(Park et al., 2008; Akeboshi et al., 2009). As an alternative strategy to overcome this limitation, chemical or chemoenzymatic methods have been developed to link M6P-oligosaccharides to the enzymes (Priyanka et al., 2016; Yamaguchi et al., 2016; Zhu et al., 2004, 2009; Zhou et al., 2011; Liu et al., 2011, 2013). Despite recent advances made in methods for preparation of glycoproteins (Unverzagt and Kajihara, 2013; Wang and Amin, 2014; van Kasteren et al., 2007; Fairbanks, 2017), production of multiple M6P-appended lysosomal enzymes still remains a herculean task. Although they are in great demand in studies and ultimate treatment of LSDs, the availability of efficient methods for preparation of multi-M6P-containing lysosomal enzymes is limited.

The genetic code expansion technique developed by Schultz and coworkers enables site-specific incorporation of unnatural amino acids (UAAs) into proteins *in vitro* and *in vivo* (Noren et al., 1989; Wang et al., 2001). This technology is a powerful tool in protein engineering owing to its technical simplicity and applicability to many proteins. To date, by using this method a variety of UAAs have been successfully incorporated into proteins with high efficiencies and specificities (Liu and Schultz, 2010; Lang and Chin, 2014; Zhang et al., 2015). In an effort to prepare multi-M6P-containing enzymes, we employed a genetic code expansion technique together with click chemistry (copper(I)-catalyzed cycloaddition reaction).

As described below, multiple M6P-appended β -N-acetylhexosaminidases were generated with a high efficiency by click chemistry between multiple M6P-appended N₃-peptides and enzymes possessing an alkynylated UAA. We also developed cell-permeable, lysosome-targeting, near-infrared (NIR)-based fluorogenic substrates for (real-time) monitoring of β -N-acetylhexosaminidase activity in lysosomes of live cells. The results of studies conducted in this effort showed that the β -N-acetylhexosaminidase bearing three M6P residues was efficiently internalized into patient-derived primary cells and was then trafficked to lysosomes. Very intriguingly, the glycoengineered enzyme removed the GM2-ganglioside accumulated in lysosomes of diseased cells in which lysosomal β -N-acetylhexosaminidase is deficient. The finding indicates that the M6P-enzyme conjugate can be used to restore β -N-acetylhexosaminidase activity in diseased cells.

RESULTS

Design and Synthesis of Multiple M6P-Appended N₃-Peptides

Tay-Sachs (TS) disease and Sandhoff disease (SD) are a class of LSDs that result from deficient activity of β -N-acetylhexosaminidase, which mainly catalyzes cleavage of GM2-ganglioside in lysosomes to form GM3-ganglioside and GalNAc (Figure 1A) (Tsuji et al., 2011). Inherited deficiencies of these enzymes lead to abnormal accumulation of GM2-ganglioside in lysosomes. In an investigation aimed at developing a strategy to remove GM2-ganglioside stored in lysosomes of diseased cells, glycoengineered N-acetylhexosaminidases were produced by using a combination of the genetic code expansion technique with click chemistry.

Previous studies showed that the CI-M6PR interacts with monovalent M6P relatively weakly ($K_d = 7 \mu\text{M}$) but with

glycoproteins containing multiple M6P residues strongly ($K_d = 2\text{--}20 \text{ nM}$) through multivalent interactions (Tong et al., 1989). Guided by these findings, we prepared multi-M6P-appended β -N-acetylhexosaminidases by click chemistry between N₃-peptides containing both multiple M6P and rhodamine (Rh) dye and alkyne-bearing enzymes, obtained by using the genetic code expansion technique. The fluorescent Rh dye was included in these substances so that entry of the derived glycopeptides and their conjugated enzymes into cells could be readily monitored.

For this purpose, the multiple M6P-appended N₃-peptides (N₃-(M6P)₁-Rh, N₃-(M6P)₂-Rh, N₃-(M6P)₃-Rh and N₃-(M6P)₆-Rh), containing varied numbers (one, two, three, and six) of M6P residues and an N-terminal azide moiety, were synthesized by using the general pathway delineated in Figure 1B. Briefly, peptides containing different numbers (one, two, three, and six) of alkyne units were assembled on a solid support, using HCTU as a coupling reagent, followed by subsequent coupling of the rhodamine derivative, Rh-Pip-CO₂H, to the side chain of the lysine residue (Hood et al., 2008; Tian et al., 2011). The *t*-butyl protected M6P derivative, N₃-M6P(*t*-Bu)₂, was then conjugated to alkyne-containing peptides on the solid support by click chemistry (Hoogendoorn et al., 2014). N₃-CO₂H linker was coupled to the N terminus of M6P-appended peptides. Finally, N₃-(M6P)_n-Rh peptides ($n = 1, 2, 3$, and 6) were obtained by removal of *t*-butyl groups for phosphate protection and cleavage from the resin under acidic conditions. The synthesized multiple M6P-appended N₃-peptides were analyzed by using reversed-phase high-pressure liquid chromatography (HPLC) and mass spectrometry (MS).

Internalization of M6P-Appended N₃-Peptides into Lysosomes

Prior to conjugation of the M6P-appended N₃-peptides to the alkyne-containing enzyme, the effect of the number of M6P residues attached to peptides on internalization into cells was examined. In this study, time-dependent internalization of peptides into C2C12 myoblasts, which express M6PR on their surfaces (Gundry et al., 2009), was initially investigated by incubating the cells with N₃-(M6P)_n-Rh peptides ($n = 1, 2, 3, 6$) for various time intervals (1–4 hr). The cells were then treated with LysoTracker Green to determine whether the peptides enter lysosomes. The results of cell image analysis using confocal fluorescence microscopy showed that N₃-(M6P)₃-Rh and N₃-(M6P)₆-Rh, containing three and six M6P residues, entered cells more efficiently than peptides bearing less than three M6P residues (N₃-(M6P)₁-Rh and N₃-(M6P)₂-Rh) (Figure 2A). The findings suggest that at least three M6P residues are necessary for efficient internalization of the peptides into cells. It was also found that entry of M6P-appended peptides into cells reached nearly saturation after 2 hr. Importantly, the peptides migrated exclusively to lysosomes, as judged by the observation of a strong co-localization of rhodamine attached to peptides with LysoTracker Green.

Next, we evaluated whether M6P-appended peptides are trafficked to lysosomes of diseased cells. In this study, TS and SD patient-derived primary fibroblasts (TS and SD fibroblasts) that lack endogenous lysosomal β -N-acetylhexosaminidase, along with C2C12 and NIH3T3 cell lines as a control, were treated with N₃-(M6P)₃-Rh for 4 hr in the absence and



(A) Alkyne-containing β -N-acetylhexosaminidase generated by a genetic code expansion technique is coupled to N₃-peptides containing multiple M6P residues by click chemistry. The glycoengineered enzyme is internalized into diseased cells via M6PR-mediated endocytosis and is then translocated to the lysosome where it catalyzes cleavage of GM2-ganglioside into GM3-ganglioside and GalNAc.

(B) Synthesis of N₃-peptides containing varied numbers of M6P residues. Reagents and conditions: (a) Fmoc-Lys(Mtt)-OH, HCTU, DIEA, DMF; (b) 20% piperidine in DMF; (c) Fmoc- β -Ala, HCTU, DIEA, DMF; (d) 10% TFA in CH₂Cl₂; (e) Rh-Pip-CO₂H, HCTU, DIEA, DMF; (f) N₃-M6P(t-Bu)₂, sodium ascorbate, CuSO₄, DMF, H₂O; (g) N₃-CO₂H linker, HCTU, DIEA, DMF; (h) TFA, TIS, H₂O. Mtt, 4-methyltrityl; HCTU, O-(6-chlorobenzotriazol-1-yl)-N,N,N',N'-tetramethyluronium hexafluorophosphate; DIEA, diisopropylethylamine; DMF, N,N-dimethylformamide; TFA, trifluoroacetic acid; TIS, triisopropylsilane.

lines (Figures 2B and S1). However, when either free M6P or NaN_3 was present in cells, the internalization efficiency of the peptide into lysosomes was greatly attenuated (Figure S1), indicating that the peptide enters cells via M6PR-mediated endocytosis.

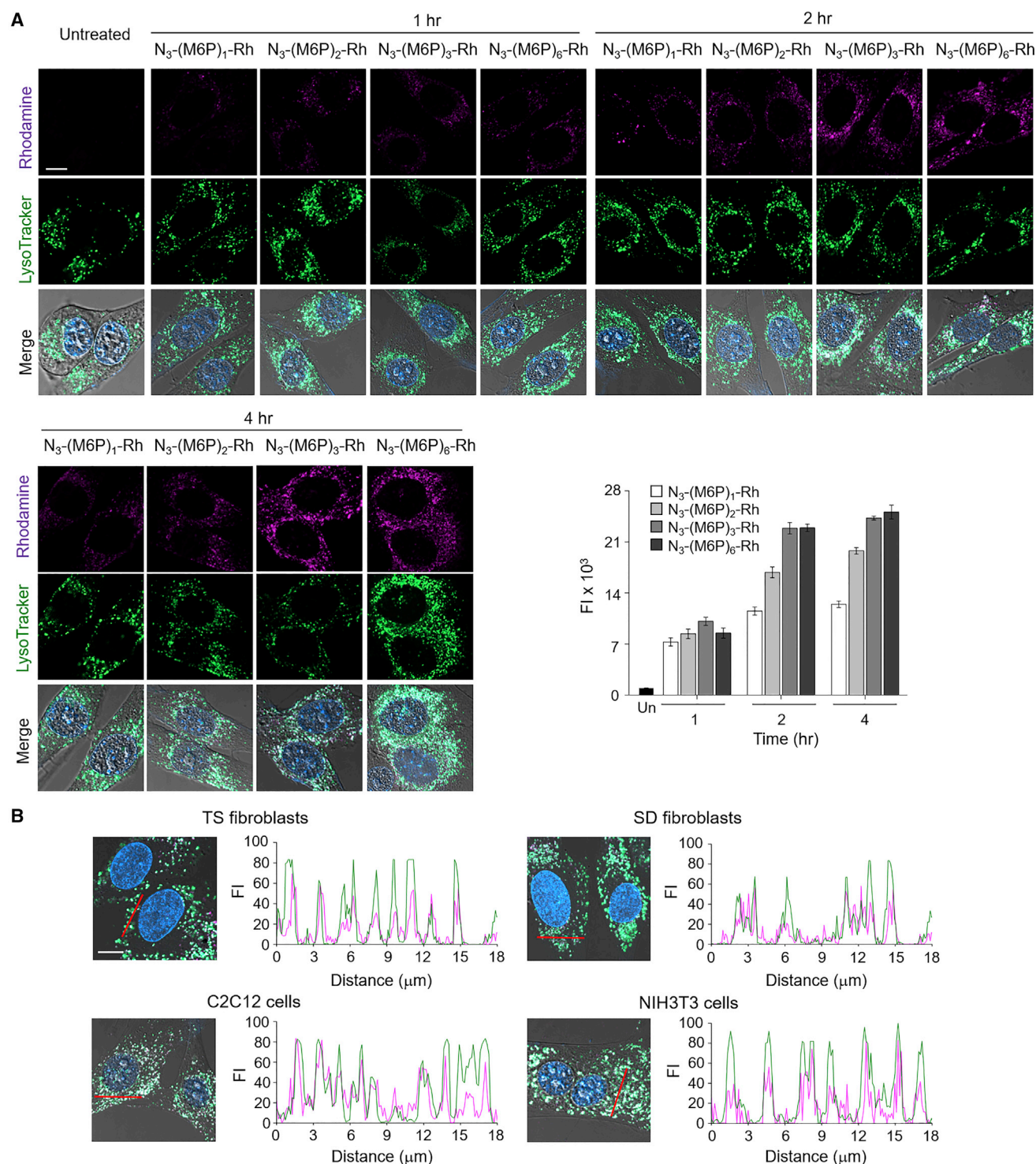


Figure 2. Internalization of M6P-Appended Peptides into Cells

(A) C2C12 cells were incubated with 10 μ M of an indicated peptide (N_3 -(M6P)_n-Rh, n = 1, 2, 3, 6) during various time intervals (1–4 hr). After washing, cells were treated with LysoTracker Green and then Hoechst 33342. Cell images were obtained by using confocal fluorescence microscopy (scale bar, 10 μ m). Merged cell images of rhodamine (scarlet), LysoTracker (green) and Hoechst 33342 (blue) are shown. λ_{ex} = 405 nm for Hoechst 33342, 488 nm for LysoTracker Green and 543 nm for rhodamine. Graphs show fluorescence intensity (FI) of rhodamine of N_3 -(M6P)_n-Rh in cells, which was analyzed by using the ZEN software. Un, untreated. Mean $\pm$ SD, n = 3.

(legend continued on next page)

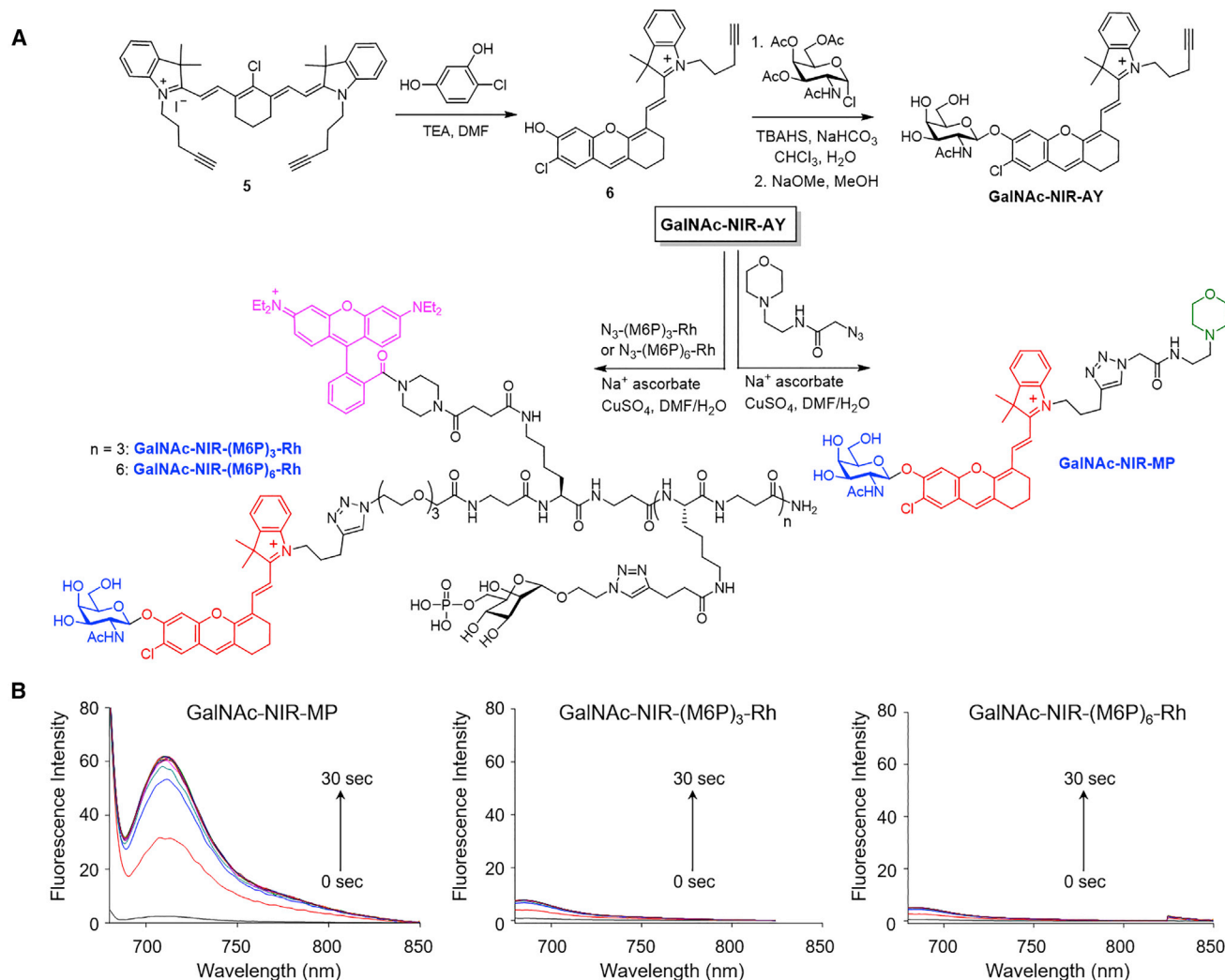


Figure 3. Synthesis of NIR-Based Fluorogenic Substrates and their Time-Dependent Responses to β -N-Acetylhexosaminidase

(A) The alkyne-bearing NIR probe **6** was glycosylated with α -GalNAc(OAc)₃-Cl to afford GalNAc-NIR-AY after removal of O-acetyl groups. The resulting product was coupled to an azide-containing morpholine derivative to give GalNAc-NIR-MP by click chemistry. In addition, GalNAc-NIR-AY was conjugated with N₃-(M6P)₃-Rh and N₃-(M6P)₆-Rh under click chemistry conditions to generate GalNAc-NIR-(M6P)₃-Rh and GalNAc-NIR-(M6P)₆-Rh, respectively.

(B) Each fluorogenic substrate (10 μ M, 50 mM phosphate buffer containing 1% DMSO, pH 5.0) was incubated with β -N-acetylhexosaminidase (50 nM) at 37°C. Changes in fluorescence intensity of each substrate were determined as a function of time by using a fluorometer (λ_{ex} = 670 nm).

Detection of β -N-Acetylhexosaminidase Activity in Lysosomes Using Fluorogenic Substrates

Cell-permeable, lysosome-targeting, fluorogenic substrates for detecting β -N-acetylhexosaminidase activity in live cells have not been developed to date. Owing to the need to measure activity of the supplemental glycoengineered enzyme in lysosomes of diseased cells, NIR-based fluorogenic substrates (GalNAc-NIR-MP, GalNAc-NIR-(M6P)₃-Rh, and GalNAc-NIR-(M6P)₆-Rh) for β -N-acetylhexosaminidase were synthesized according to the pathway shown in Figure 3A (Hyun et al., 2018). Morpholine

(MP) and multiple M6P residues were attached to the probes to target lysosomes.

The spectral properties of NIR fluorogenic substrates were initially examined. The maximum absorbance of the non-glycosylated probe NIR-MP in aqueous buffer at pH 5.0, at which lysosomal enzyme catalysis occurs optimally, was higher than that of its glycosylated form (GalNAc-NIR-MP), likely a result of partial deprotonation of the OH group (pK_a = approximately 4.5) of NIR-MP (Figure S2A) (Yuan et al., 2012; Zhang et al., 2017). However, GalNAc-NIR-MP showed almost no fluorescence, while

(B) TS fibroblasts, SD fibroblasts, C2C12 cells, and NIH3T3 cells were separately incubated with 10 μ M N₃-(M6P)₃-Rh for 4 hr. After washing, cells were treated with LysoTracker Green and then Hoechst 33342. Cell images were obtained using confocal fluorescence microscopy (scale bar, 10 μ m). Shown are merged cell images of rhodamine (scarlet), LysoTracker (green), and Hoechst 33342 (blue). Graphs show fluorescence intensity (FI) of rhodamine (scarlet) of N₃-(M6P)₃-Rh and LysoTracker (green) at positions along the red line across the cell. Experiments were repeated three times, giving similar results (see Figure S1 for more cell images).

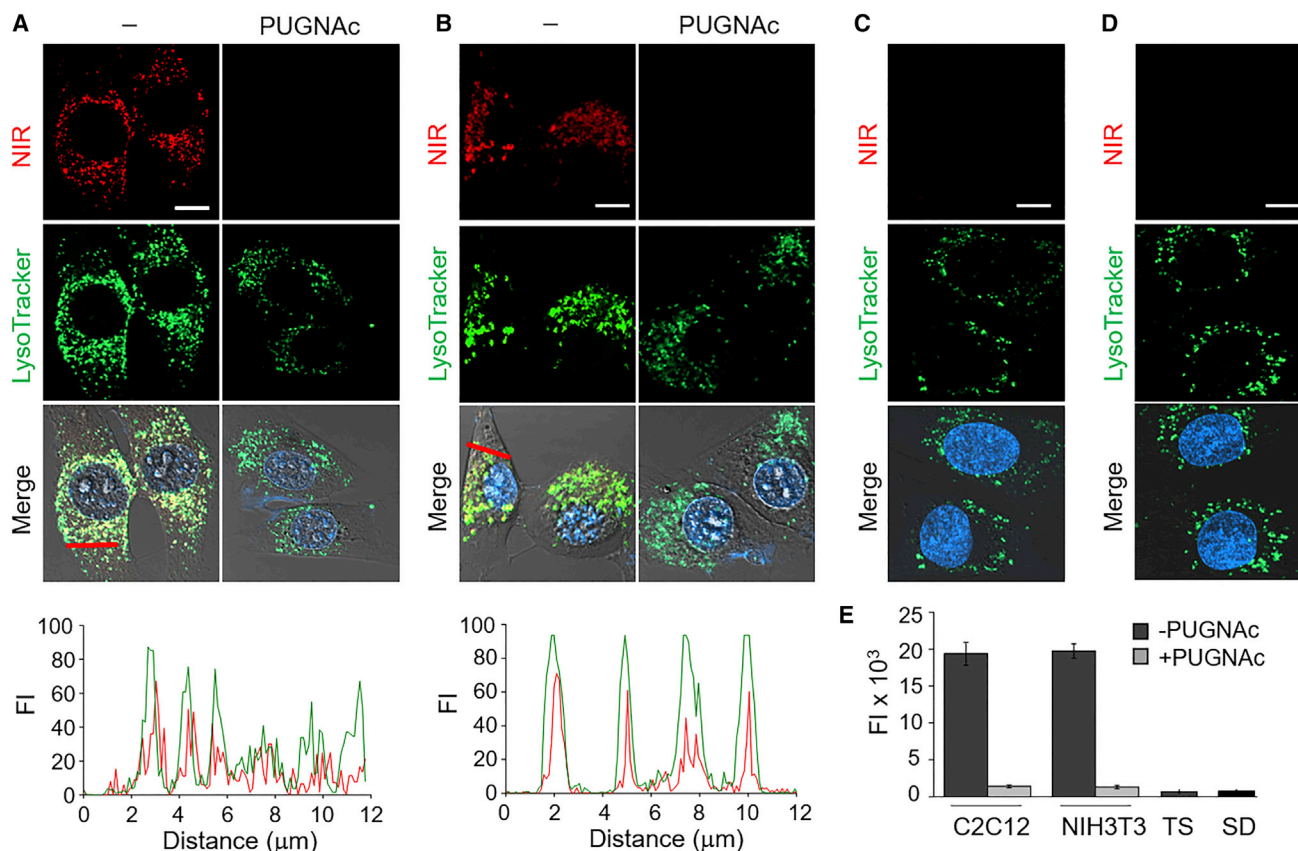


Figure 4. Detection of β -N-Acetylhexosaminidase Activity in Cells Using GalNAc-NIR-MP

(A and B) (A) C2C12 and (B) NIH3T3 cells were incubated for 4 hr with 20 μ M GalNAc-NIR-MP in the (left) absence and (right) presence of 50 μ M PUGNac. After washing, cells were treated with LysoTracker Green and then Hoechst 33342. Cell images were obtained using confocal fluorescence microscopy (scale bar, 10 μ m). Fluorescence intensity (FI) of the NIR probe in cells was analyzed by using the ZEN software. Bottom images: merged cell images of an NIR probe (red), LysoTracker (green), and Hoechst 33342 (blue). λ_{ex} = 405 nm for Hoechst 33342, 488 nm for LysoTracker, and 640 nm for an NIR probe. Graphs in (A) and (B) are fluorescence intensities of an NIR probe (red) arising from GalNAc-NIR-MP and LysoTracker (green) at positions along the red line across each cell. Experiments were repeated three times, giving similar results.

(C and D) (C) TS and (D) SD fibroblasts were incubated with 20 μ M GalNAc-NIR-MP for 4 hr. After washing, cells were treated with LysoTracker Green and then Hoechst 33342. Cell images were obtained using confocal fluorescence microscopy (scale bar, 10 μ m).

(E) Graphs show fluorescence intensity of the NIR probe in (A)–(D) in the absence and presence of PUGNac (mean $\pm$ SD, n = 3).

NIR-MP was highly emissive upon excitation at 670 nm (Figure S2B). The greatly different fluorescence properties of the non-glycosylated and glycosylated NIR probes suggest that NIR fluorogenic substrates would be useful for sensitively monitoring β -N-acetylhexosaminidase activity.

Indeed, treatment of GalNAc-NIR-MP with β -N-acetylhexosaminidase resulted in a large, rapid, and time-dependent increase in fluorescence intensity (Figure 3B). However, the glycopeptide-based NIR probes, GalNAc-NIR-(M6P)₃-Rh and GalNAc-NIR-(M6P)₆-Rh, generated slightly increased fluorescence signals in response to treatment of β -N-acetylhexosaminidase, presumably owing to their unfavorably large size. We also observed that O-(2-acetamido-2-deoxy-D-glucopyranosylidene)amino-N-phenylcarbamate (PUGNac) acted as an inhibitor of β -N-acetylhexosaminidase (Figure S2C) (Kim et al., 2006).

The ability of fluorogenic substrates to detect β -N-acetylhexosaminidase activity in lysosomes was investigated next. C2C12 cells were incubated with GalNAc-NIR-MP and GalNAc-NIR-(M6P)₆-Rh. The results of cell image analysis showed that NIR

fluorescence of cells treated with GalNAc-NIR-MP greatly increased after incubation for 2 hr, and the signal overlapped with the fluorescence of LysoTracker Green (Figures S2D and S2E). Contrary to observations made using GalNAc-NIR-MP, NIR fluorescence in cells treated with GalNAc-NIR-(M6P)₆-Rh was weak, even though rhodamine fluorescence was relatively strong (Figure S2D). The findings indicated that GalNAc-NIR-(M6P)₆-Rh was efficiently translocated to the lysosome but was a poor substrate for the lysosomal β -N-acetylhexosaminidase.

To examine the selective response of GalNAc-NIR-MP to lysosomal β -N-acetylhexosaminidase, C2C12 and NIH3T3 cells were incubated with the probe for 4 hr in the absence and presence of PUGNac while monitoring lysosomal β -N-acetylhexosaminidase activities. As shown in Figures 4A and 4B, both cells displayed strong fluorescence of an NIR probe at locations that overlapped with the fluorescence of LysoTracker Green. However, the NIR fluorescence signals emanating from the treated cells were greatly reduced when PUGNac was present. The finding indicates that GalNAc-NIR-MP responds to lysosomal

β -N-acetylhexosaminidase selectively. As expected, SD and TS fibroblasts did not exhibit fluorescence of an NIR probe when incubated with GalNAc-NIR-MP (Figures 4C and 4D).

Experiments were also carried out to determine if GalNAc-NIR-MP can be utilized for real-time monitoring of β -N-acetylhexosaminidase activity in live cells. To attain this objective, C2C12 cells pre-treated with LysoTracker Green were incubated with GalNAc-NIR-MP in the absence and presence of PUGNAc. Cell images were obtained using confocal fluorescence microscopy every 5 min for a 2 hr incubation period without washing. The results of cell image analysis showed that an NIR fluorescence signal arising from GalNAc-NIR-MP, which overlapped with the fluorescence of LysoTracker Green, was enhanced following a 60 min incubation time and reached maximum after 100 min (Figures S3A and S3C). However, the NIR fluorescence signal arising from GalNAc-NIR-MP was almost completely abrogated in the presence of PUGNAc (Figures S3B and S3C). Collectively, the findings support the conclusion that GalNAc-NIR-MP has the ability to detect lysosomal β -N-acetylhexosaminidase activity in lysosomes of live cells in a selective and sensitive manner.

Preparation of M6P-Appended Enzymes

In the next phase of this investigation, we prepared multi-M6P-conjugated β -N-acetylhexosaminidases by coupling enzymes containing an alkyne-linked lysine (Lys(PA)) (Nguyen et al., 2009) with N_3 -(M6P)₃-Rh under click chemistry conditions. Because N_3 -(M6P)₃-Rh and N_3 -(M6P)₆-Rh were internalized into lysosomes to a similar degree, the former was selected for ligation reactions with the UAA-incorporated enzyme. Among UAAs that have been utilized for site-specific protein conjugation (Lang and Chin, 2014), Lys(PA) was chosen because other UAAs containing strained alkynes/alkenes and tetrazines are synthetically challenging and unstable under certain conditions, and Cu(I)-catalyzed click reaction is efficient enough for this study. The required alkyne-containing β -N-acetylhexosaminidases were expressed in *Escherichia coli* by using the genetic code expansion technique. Residues Lys432 and Lys561 of β -N-acetylhexosaminidase were chosen as sites for genetic incorporation of Lys(PA) since they are exposed to solvent and are located in loop regions remote from the active site (Figure 5A) (Jiang et al., 2011).

With this in mind, an amber (TAG) mutation was introduced into codons for Lys432 and Lys561 in the β -N-acetylhexosaminidase gene. Because a His₆-tag widely used for affinity purification is known to negatively affect click chemistry (Presolski et al., 2011), the C terminus of the enzyme was designed to contain a Strep-tag for purification. The two mutants, Hex-K432Lys(PA) and Hex-K561Lys(PA), possessing Lys(PA) at positions 432 and 561, were expressed in the *E. coli* DH10 β strain, which was transformed with a plasmid containing pyrrolysyl-tRNA(PylT) and pyrrolysyl-tRNA synthetase (PylRS) from *Methanobacterium mazei* (Lee et al., 2013) in culture media containing 3 mM Lys(PA). The produced mutants were subjected to Strep-Tactin affinity chromatography to give pure proteins (Hex-K432Lys(PA), 7.2 mg/L; Hex-K561Lys(PA), 9.4 mg/L). The results of SDS-PAGE analysis showed that the full-length mutant enzymes were expressed in the presence of Lys(PA) but not in its absence (Figure 5B). A wild-type (WT) enzyme (WT-Hex) was

also expressed and purified for use in control experiments (17.3 mg/L). The incorporation of Lys(PA) at the selected positions in both mutants was confirmed by MS analysis (Figures 5C and 5D). No incorporation of any natural amino acid was observed in the mutant enzymes.

To ligate the alkyne-containing β -N-acetylhexosaminidases with multiple M6P-appended N_3 -peptides, Hex-K432Lys(PA) and Hex-K561Lys(PA), along with WT-Hex as a control, were coupled with N_3 -(M6P)₃-Rh by click chemistry for 1–3 hr at 25°C (Besanceney-Webler et al., 2011). After quenching the reactions by addition of EDTA, the mixtures were analyzed by SDS-PAGE. It was found that no ligated enzyme was formed when WT-Hex and N_3 -(M6P)₃-Rh were subjected to click chemistry, indicating that the Lys(PA) unit is required for ligation (Figure S4A). However, the ligated enzyme Hex-K561Lys-(M6P)₃-Rh was successfully generated with a very high efficiency (more than 95%) after a coupling reaction of Hex-K561Lys(PA) with N_3 -(M6P)₃-Rh for 1 hr (Figures S4A and S4C). In marked contrast, the ligated enzyme Hex-K432Lys-(M6P)₃-Rh was produced only with a very low efficiency even after a 3 hr coupling reaction of Hex-K561Lys(PA) with N_3 -(M6P)₃-Rh (Figure S4B). Because K432 is located in a loop region relatively close to two helical structures of the enzyme (Figure 5A), this site might not be easily accessed by N_3 -(M6P)₃-Rh for conjugation. However, K561 is located in a loop region relatively remote from helical structures of the enzyme and, as a result, the ligated enzyme Hex-K561Lys-(M6P)₃-Rh was successfully produced by click chemistry.

Kinetic studies were conducted to assess whether introduction of the multiple M6P-appended peptide into the enzyme affects its catalytic activity. The rates of increases in the NIR fluorescence intensity of GalNAc-NIR-MP were determined at various concentrations after treatment of WT-Hex, Hex-K561Lys(PA), and Hex-K561Lys-(M6P)₃-Rh. The results showed that WT-Hex and Hex-K561Lys(PA) had very similar k_{cat} and K_m values (Figures 5E and S4D), indicating that mutation of Lys to Lys(PA) does not affect catalytic activity. In addition, the ligated enzyme Hex-K561Lys-(M6P)₃-Rh had similar catalytic activity to WT-Hex, showing that the enzyme activity is not affected by the presence of the multiple M6P-appended peptide. We also evaluated the activity of the glycoengineered enzyme toward cleavage of GM2-oligosaccharide to form GM3-oligosaccharide. The results of thin-layer chromatography analysis of enzymatic reaction mixtures showed that GM2-oligosaccharide was degraded by Hex-K561Lys-(M6P)₃-Rh with a similar efficiency to those of WT-Hex and Hex-K561Lys(PA) (Figure S4E).

Degradation of GM2-Ganglioside Stored in Lysosomes of Diseased Cells by Supplemental Hex-K561Lys-(M6P)₃-Rh

The ability of the glycoengineered β -N-acetylhexosaminidase to enter cells was evaluated by incubating SD and TS fibroblasts with Hex-K561Lys-(M6P)₃-Rh for various times (0–14 hr). For comparison purposes, the cells were also treated under the same conditions with rhodamine-labeled, commercialized human β -hexosaminidase B (Rh-HEXB), which was produced in mammalian cells. In addition, cells were incubated for 14 hr with a Cy3-labeled WT-Hex (Cy3-WT-Hex), which

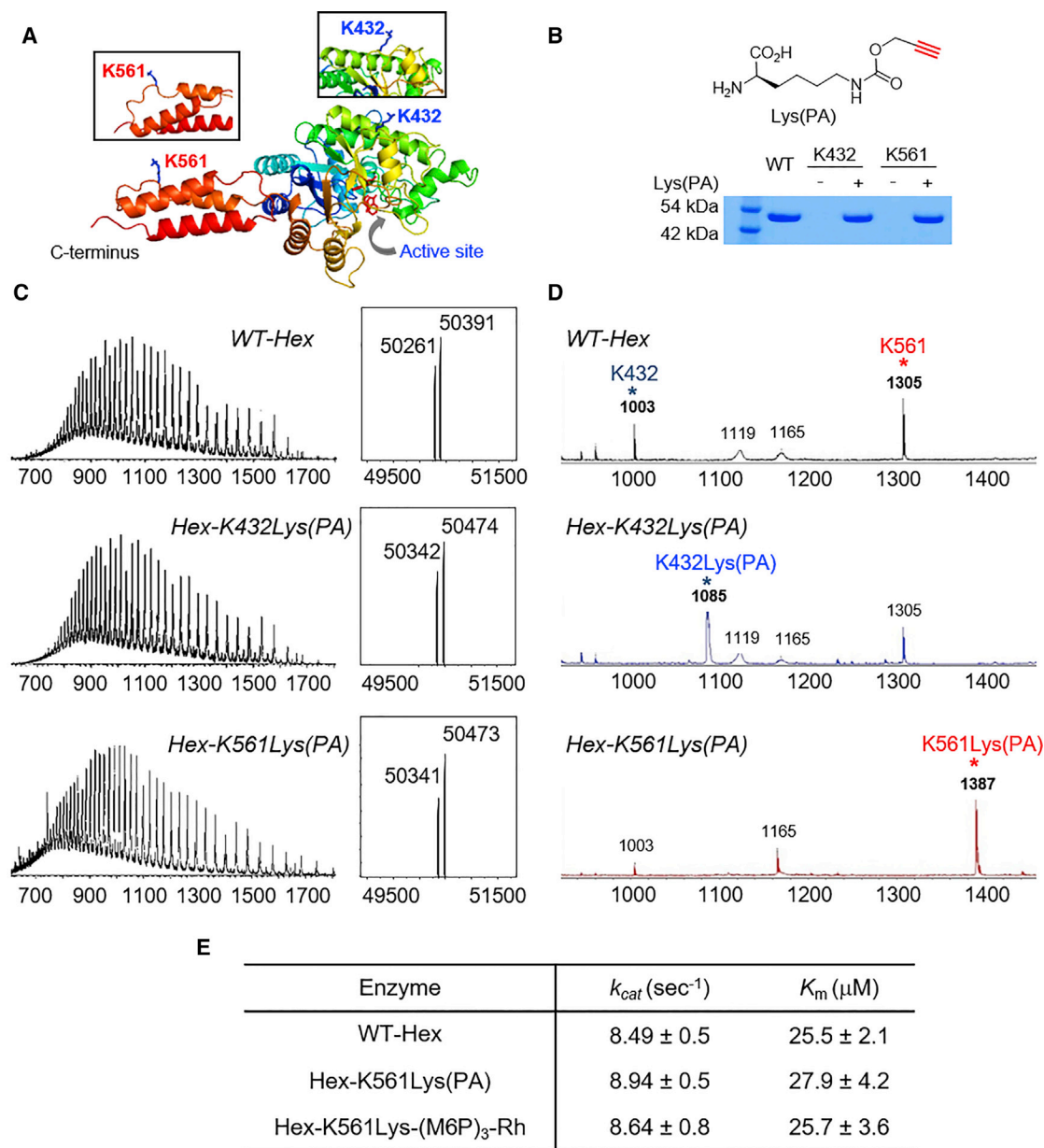


Figure 5. Genetic Incorporation of Lys(PA) into β-N-Acetylhexosaminidase and Enzyme Kinetics

(A) The two mutation sites, Lys432 and Lys561, for incorporation of Lys(PA) into β-N-acetylhexosaminidase are shown in the X-ray crystal structure (PDB: 3RPM). Rotated structures are shown in the box to more clearly visualize the mutated sites.

(B) SDS-PAGE analysis of WT-Hex and two mutants, Hex-K432Lys(PA) and Hex-K561Lys(PA), expressed in the absence and presence of Lys(PA).

(C) Electrospray ionization MS (ESI-MS) of WT-Hex, Hex-K432Lys(PA), and Hex-K561Lys(PA). Boxes show the deconvoluted spectra. WT-Hex: expected mass, 50,391 Da, 50,260 Da (-Met); observed mass, 50,391 Da, 50,261 Da (-Met). Hex-K432Lys(PA): expected mass, 50,473 Da, 50,342 Da (-Met); observed mass, 50,474 Da, 50,342 Da (-Met). Hex-K561Lys(PA): expected mass, 50,473 Da, 50,342 Da (-Met); observed mass, 50,473 Da, 50,341 Da (-Met).

(D) MALDI-TOF MS of peptide fragments after Glu-C digestion of WT, Hex-K432Lys(PA), and Hex-K561Lys(PA). Peptide K432 (residues 425–433; TSFGSFDK) and peptide K561 (residues 559–569; VPKDLNRYTAE) represent the Glu-C digested peptide fragments containing residues 432 and 561, respectively. Peptide K432Lys(PA) and peptide K561Lys(PA) represent the peptides containing Lys(PA) at positions 432 and 561, respectively.

(E) Kinetic parameters of WT-Hex, Hex-K561Lys(PA), and Hex-K561Lys-(M6P)₃-Rh. The parameters were obtained from the results of enzyme kinetics shown in Figure S4D (mean ± SD, n = 3).

lacks M6P residues, as a control. The internalization efficiency of the enzyme into cells was determined by measuring fluorescence signals arising from the rhodamine dye linked to the enzyme. The observation that rhodamine fluorescence occur-

ring in cells overlapped with the fluorescence of LysoTracker Green showed that Hex-K561Lys-(M6P)₃-Rh and Rh-HEXB were internalized into cells and then migrated to lysosomes (Figures S5A and S5B). In marked contrast, the cells treated

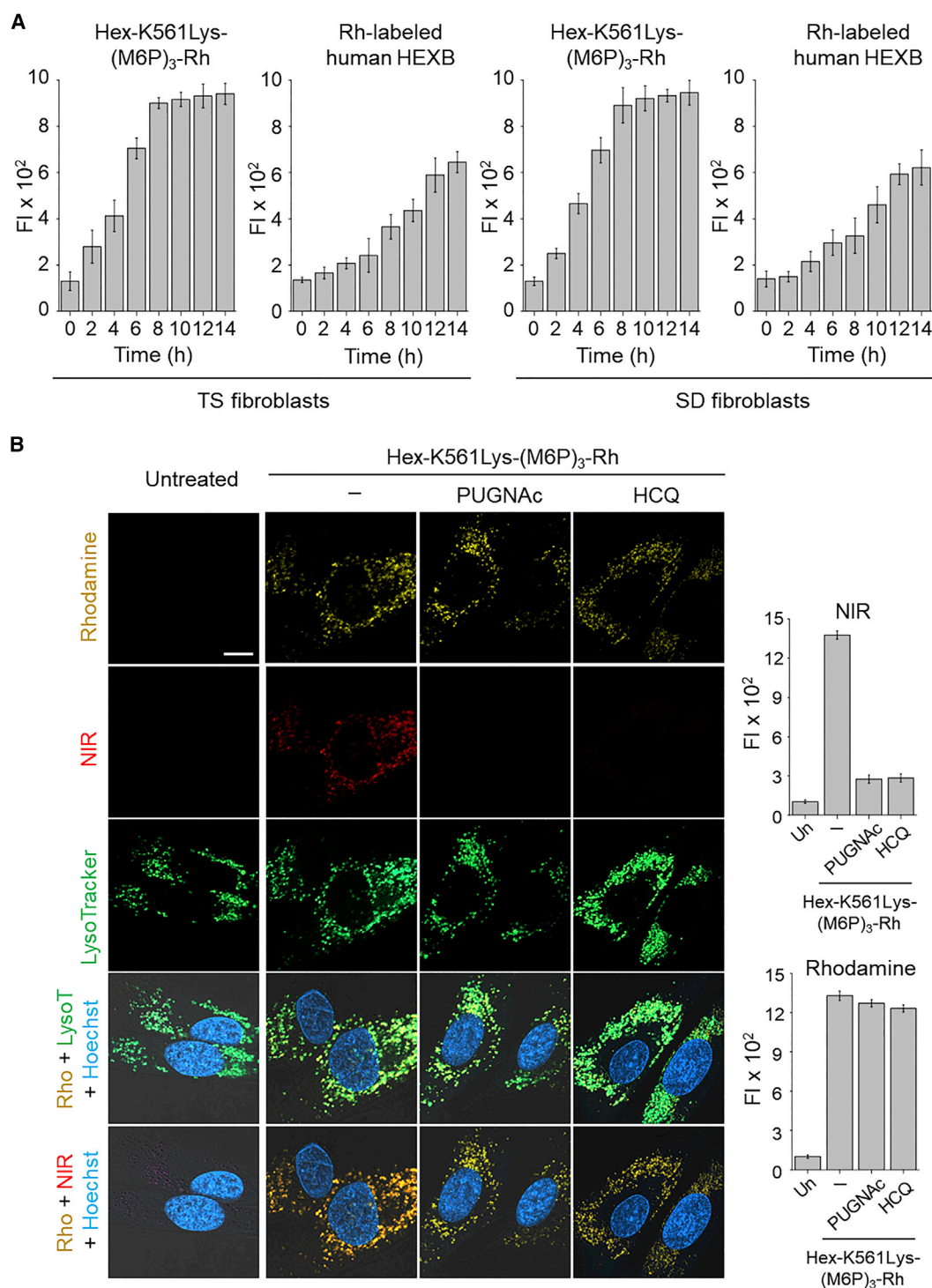


Figure 6. Internalization of Hex-K561Lys-(M6P)₃-Rh into Diseased Cells and Detection of Its Activity Using an NIR Probe

(A) TS and SD fibroblasts were incubated for various time periods with 50 nM Hex-K561Lys-(M6P)₃-Rh or 50 nM rhodamine-labeled commercialized human HEXB which was produced in mammalian cells. Cell images were obtained by using confocal fluorescence microscopy (Figure S5). Fluorescence intensities of rhodamine linked to enzymes in cells were analyzed by using the ZEN software.

(B) TS fibroblasts were treated for 10 hr with 100 nM Hex-K561Lys-(M6P)₃-Rh in the absence and presence of either PUGNAc (50 μ M) or HCQ (10 μ M). After washing, cells were incubated with 20 μ M GaINac-NIR-MP for 4 hr, followed by treatment with LysoTracker Green and Hoechst 33342. Cell images were obtained using confocal fluorescence microscopy (scale bar, 10 μ m). Fluorescence intensities of NIR and rhodamine probes in cells were analyzed by using the ZEN software. $\lambda_{\text{ex}} = 405$ nm for Hoechst 33342 (blue), 488 nm for LysoTracker (green), 543 nm for rhodamine (yellow), and 640 nm for an NIR probe (red). Graphs show fluorescence intensities of (upper) NIR and (lower) rhodamine probes in cells. FI, fluorescence intensity; Un, untreated.

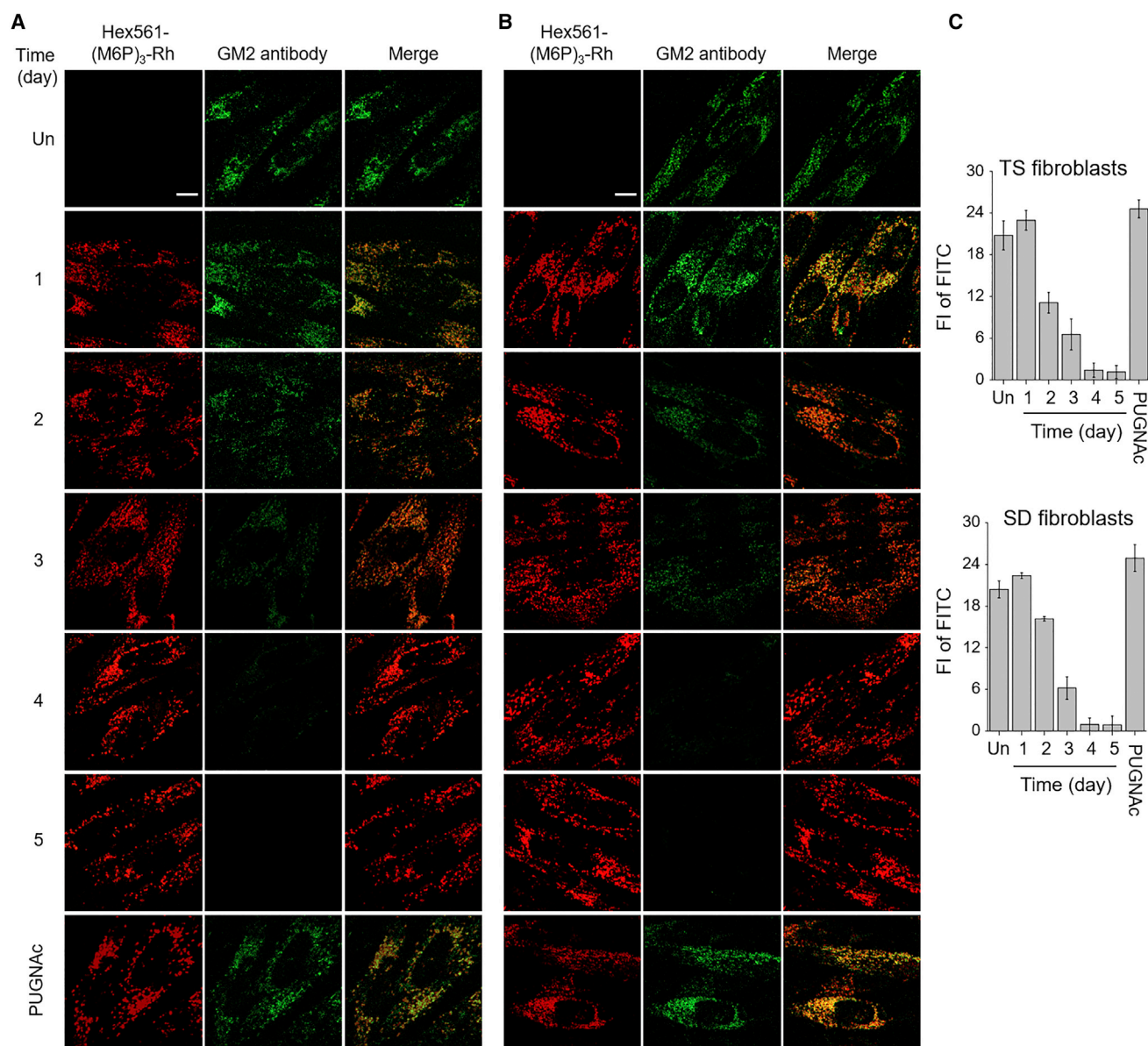


Figure 7. Degradation of GM2-ganglioside Stored in Lysosomes of Diseased Cells by Supplemental Hex-K561Lys-(M6P)₃-Rh

(A and B) (A) TS and (B) SD fibroblasts were incubated with 200 nM Hex-K561Lys-(M6P)₃-Rh for different time intervals. In addition, SD fibroblasts were incubated for 5 days with 200 nM Hex-K561Lys-(M6P)₃-Rh in the presence of 50 μ M PUGNAC. The treated cells were fixed at the indicated time and GM2-ganglioside was detected after treatment with GM2-ganglioside antibody followed by treatment with fluorescein isothiocyanate (FITC)-labeled secondary antibody. Cell images were obtained using confocal microscopy (scale bar, 10 μ m). Fluorescence intensity of FITC in cells was analyzed by using the ZEN software. λ_{ex} = 488 nm for GM2 antibody (green) and 543 nm for rhodamine (red). Un, untreated

(C) Fluorescence intensity (FI) of FITC of antibody shown in (A) and (B).

with Cy3-WT-Hex exhibited very weak fluorescence of Cy3 (Figures S5A and S5B), indicating that M6P residues are essential for internalization of the enzyme into cells. Importantly, Hex-K561Lys-(M6P)₃-Rh entered cells more rapidly than did Rh-HEXB (Figures S5 and 6A), showing that the homogeneous multi-M6P-containing enzyme, generated by combining genetic code expansion with bioorthogonal ligation techniques, has a better ability to enter cells than Rh-HEXB, which was produced in mammalian cells and thus might contain the incomplete M6P glycosylation.

Next, to determine if Hex-K561Lys-(M6P)₃-Rh is catalytically active in lysosomes, SD and TS fibroblasts were incubated with the enzyme in the absence and presence of either PUGNAC or hydroxychloroquine (HCQ). It is known that HCQ elevates the lysosomal pH, causing disruption of lysosomal enzyme activities, which are optimal at low pH (pH $\leq$ 5) (Xie and Klionsky, 2007; Homewood et al., 1972). The treated cells were then incubated with GalNAc-NIR-MP to detect enzyme activity in lysosomes. The cells treated with Hex-K561Lys-(M6P)₃-Rh displayed fluorescence signals arising from both rhodamine and

the NIR probe, which co-localized with LysoTracker fluorescence (Figures 6B and S6). However, when either PUGNAc or HCQ was present, a very low fluorescence signal of an NIR probe was produced but an intense emission from rhodamine was observed in cells (Figures 6B and S6). This finding indicates that although the enzyme migrates to the lysosome (high rhodamine fluorescence in lysosomes), its activity is greatly attenuated in the presence of PUGNAc or HCQ (low fluorescence of an NIR probe in lysosomes). The overall findings serve as evidence to support the conclusion that the catalytically active M6P-appended enzyme is delivered to lysosomes of diseased cells.

Finally, we investigated the effect of Hex-K561Lys-(M6P)₃-Rh on the degradation of GM2-ganglioside accumulated in diseased cells by employing an immunocytochemistry-based assay with GM2-ganglioside antibody (Ohsawa et al., 2005; Itakura et al., 2006). In this study, SD and TS fibroblasts were treated with Hex-K561Lys-(M6P)₃-Rh for various times and then immunostained with GM2-ganglioside antibody. Analysis of cell images of the untreated SD and TS fibroblasts showed that GM2-ganglioside accumulates in the diseased cells (Figures 7 and S7). Importantly, as the incubation time increased, the level of GM2-ganglioside gradually decreased and then almost disappeared after 4 days. In addition, fluorescence signals arising from rhodamine linked to the enzyme in lysosomes co-localized well with those of the antibody. This indicates that GM2-antibody recognizes lysosomal GM2-ganglioside and gradual degradation of GM2-ganglioside by the internalized enzyme occurs. However, GM2-ganglioside in cells treated with the enzyme was not degraded when PUGNAc was present. Taken together, the findings indicate that the glycoengineered enzyme, prepared by combining genetic code expansion and bioorthogonal ligation techniques, enter diseased cells and that it degrades GM2-ganglioside stored in lysosomes.

DISCUSSION

Lysosomal enzymes produced in mammalian cells, which are used to treat LSDs, possess heterogeneous glycans and relatively low levels of M6P modification. Therefore, it is highly desirable to create an efficient method to generate lysosomal enzymes containing multiple M6P residues. In the current study we have demonstrated that a combination of the genetic code expansion technique and click chemistry is powerful to produce catalytically active, multiple M6P-appended β -N-acetylhexosaminidase. We have also shown that the glycoengineered enzyme is efficiently internalized into cells via M6PR-mediated endocytosis and is transported to lysosomes. Importantly, the glycoengineered enzyme enters lysosomes of patient-derived primary cells more rapidly than commercialized human hexosaminidase, which is produced in mammalian cells. More importantly, the glycoengineered enzyme degraded GM2-ganglioside accumulated in lysosomes of diseased cells. We also prepared and employed a cell-permeable, lysosome-targeting, NIR fluorogenic probe for monitoring activities of endogenous and supplemental glycoengineered β -N-acetylhexosaminidases in lysosomes of live cells. In this study, we have provided an efficient method to generate catalytically active lysosomal enzymes containing multiple M6Ps. However, it should be noted that the glycoengineered enzymes produced by the present method might be immunogenic

in animals or humans; the *E. coli* system might be ineffective to express stable, properly folded human lysosomal enzymes; and, thus, more studies are needed for practical applications.

SIGNIFICANCE

A genetic code expansion technique together with bioorthogonal ligation was utilized to prepare the glycosidase conjugated with multiple M6Ps, which are critical for targeting lysosomes. The multiple M6P-conjugated glycosidase was generated with a high efficiency by using these combined techniques. Very intriguingly, the glycoengineered enzyme is translocated into lysosomes of patient-derived primary cells and removed substances accumulated in lysosomes. In addition, a cell-permeable, lysosome-targeting, NIR-based fluorogenic probe was employed to detect activities of endogenous and supplemental glycoengineered enzymes in lysosomes of live cells. Because multi-M6P-containing lysosomal enzymes can be efficiently produced by using the present coupled strategy, it is anticipated that this study aids the generation of other types of lysosomal enzymes containing multiple M6Ps.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- **KEY RESOURCES TABLE**
- **CONTACT FOR REAGENT AND RESOURCE SHARING**
- **EXPERIMENTAL MODEL AND SUBJECT DETAILS**
- **METHOD DETAILS**
 - Expression and Purification of Mutant β -N-acetylhexosaminidase Containing Lys(PA)
 - Analysis of Incorporation of Lys(PA) into Proteins by MS
 - Ligation of M6P-appended N₃-peptides to Lys(PA)-Incorporated β -N-acetylhexosaminidase by Click Chemistry
 - Enzyme Kinetics
 - Cleavage of GM2-oligosaccharide to GM3-oligosaccharide by β -N-Acetylhexosaminidase
 - General Method to Quantitate Fluorescence Intensity of Fluorescent Dyes in Cells
 - Internalization of M6P-appended Peptides into Cells
 - Internalization of Hex-K561Lys-(M6P)₃-Rh into Cells
 - Efficiency of Fluorogenic Substrates to Detect Intracellular β -N-acetylhexosaminidase Activity
 - Detection of β -N-acetylhexosaminidase Activity in Cells Using GalNAc-NIR-MP
 - Detection of Hex-K561Lys-(M6P)₃-Rh Activity Internalized into Diseased Cells Using GalNAc-NIR-MP
 - Degradation of GM2-ganglioside Accumulated in Diseased Cells by Hex-K561Lys-(M6P)₃-Rh
 - General for Synthesis
 - 2-Azidoethyl 2,3,4-tri-O-acetyl-6-O-(tert-butylidimethylsilyl)- α -D-mannopyranoside (2)
 - 2-Azidoethyl 2,3,4-tri-O-acetyl- α -D-mannopyranoside (3)

- 2-Azidoethyl 2,3,4-tri-O-acetyl-6-O-(di-tert-butylphosphoryl)- α -D-mannopyranoside (4)
- 2-Azidoethyl 6-O-(di-tert-butylphosphoryl)- α -D-mannopyranoside (N_3 -M6P(*t*-Bu)₂)
- Solid-phase Synthesis of M6P-appended N_3 -peptides
- Compound **6**
- GalNAc(OAc)₃-NIR-AY
- GalNAc-NIR-AY
- GalNAc-NIR-MP
- GalNAc-NIR-(M6P)₃ or 6-Rh
- **QUANTITATION AND STATISTICAL ANALYSIS**
 - Statistics
- **DATA AND SOFTWARE AVAILABILITY**

SUPPLEMENTAL INFORMATION

Supplemental Information includes seven figures and can be found with this article online at <https://doi.org/10.1016/j.chembiol.2018.07.011>.

ACKNOWLEDGMENTS

This study was supported financially by the National Creative Research Initiative (grant number 2010-0018272 to I.S.) and the Global Frontier Research Program (NRF-2015M3A6A8065833 to H.S.L.) in Korea.

AUTHOR CONTRIBUTIONS

I.S. designed the study and wrote the manuscript. H.S.L. helped with writing the manuscript. I.S. and H.S.L. also provided supervision of the project. J.Y.H. designed, synthesized, and characterized the compounds and performed biochemical/biological studies. S.K. conducted isolation and characterization of UAA-incorporated enzymes.

DECLARATION OF INTERESTS

The authors declare no competing financial interests.

Received: February 21, 2018

Revised: May 20, 2018

Accepted: July 25, 2018

Published: August 23, 2018

REFERENCES

- Akeboshi, H., Kasahara, Y., Tsuji, D., Itoh, K., Sakuraba, H., Chiba, Y., and Jigami, Y. (2009). Production of human beta-hexosaminidase a with highly phosphorylated N-glycans by the overexpression of the *Ogataea minuta* MNN4 gene. *Glycobiology* 19, 1002–1009.
- Besanceney-Webler, C., Jiang, H., Zheng, T., Feng, L., Soriano del Amo, D., Wang, W., Klivansky, L.M., Marlow, F.L., Liu, Y., and Wu, P. (2011). Increasing the efficacy of bioorthogonal click reactions for bioconjugation: a comparative study. *Angew. Chem. Int. Ed.* 50, 8051–8056.
- Braulke, T., and Bonifacino, J.S. (2009). Sorting of lysosomal proteins. *Biochim. Biophys. Acta* 1793, 605–614.
- Crawley, A.C., Niedzielski, K.H., Isaac, E.L., Davey, R.C., Byers, S., and Hopwood, J.J. (1997). Enzyme replacement therapy from birth in a feline model of mucopolysaccharidosis type VI. *J. Clin. Invest.* 99, 651–662.
- Dahms, N.M., Olson, L.J., and Kim, J.J. (2008). Strategies for carbohydrate recognition by the mannose 6-phosphate receptors. *Glycobiology* 18, 664–678.
- Fairbanks, A.J. (2017). The ENGases: versatile biocatalysts for the production of homogeneous N-linked glycopeptides and glycoproteins. *Chem. Soc. Rev.* 46, 5128–5146.
- Futerman, A.H., and van Meer, G. (2004). The cell biology of lysosomal storage disorders. *Nat. Rev. Mol. Cell Biol.* 5, 554–565.
- Galibert, M., Dumy, P., and Boturyn, D. (2009). One-pot approach to well-defined biomolecular assemblies by orthogonal chemoselective ligations. *Angew. Chem. Int. Ed.* 48, 2576–2579.
- Ghosh, P., Dahms, N.M., and Kornfeld, S. (2003). Mannose 6-phosphate receptors: new twists in the tale. *Nat. Rev. Mol. Cell Biol.* 4, 202–212.
- Gundry, R.L., Raginski, K., Tarasova, Y., Tchernyshyov, I., Bausch-Fluck, D., Elliott, S.T., Boheler, K.R., Van Eyk, J.E., and Wollscheid, B. (2009). The mouse C2C12 myoblast cell surface N-Linked glycoproteome. *Mol. Cell. Proteomics* 8, 2555–2569.
- He, X., Galpin, J.D., Tropak, M.B., Mahuran, D., Haselhorst, T., von Itzstein, M., Kolarich, D., Packer, N.H., Miao, Y., Jiang, L., et al. (2012). Production of active human glucocerebrosidase in seeds of *Arabidopsis thaliana* complex-glycan-deficient (cgl) plants. *Glycobiology* 22, 492–503.
- Homewood, C.A., Warhurst, D.C., Peters, W., and Baggaley, V.C. (1972). Lysosomes, pH and the anti-malarial action of chloroquine. *Nature* 235, 50–52.
- Hood, C.A., Fuentes, G., Patel, H., Page, K., Menakuru, M., and Park, J.H. (2008). Fast conventional Fmoc solid-phase peptide synthesis with HCTU. *J. Pept. Sci.* 14, 97–101.
- Hoogendoorn, S., van Puijvelde, G.H.M., Kuiper, J., van der Marel, G.A., and Overkleef, H.S. (2014). A multivalent ligand for the mannose-6-phosphate receptor for endolysosomal targeting of an activity-based probe. *Angew. Chem. Int. Ed.* 53, 10975–10978.
- Hyun, J.Y., Kang, N.R., and Shin, I. (2018). Carbohydrate microarrays containing glycosylated fluorescent probes for assessment of glycosidase activities. *Org. Lett.* 20, 1240–1243.
- Itakura, T., Kuroki, A., Ishibashi, Y., Tsuji, D., Kawashita, E., Higashine, Y., Sakuraba, H., Yamanaka, S., and Itoh, K. (2006). Inefficiency in GM2 ganglioside elimination by human lysosomal beta-hexosaminidase beta-subunit gene transfer to fibroblastic cell line derived from Sandhoff disease model mice. *Biol. Pharm. Bull.* 29, 1564–1569.
- Jeyakumar, M., Butters, T.D., Dwek, R.A., and Platt, F.M. (2002). Glycosphingolipid lysosomal storage diseases: therapy and pathogenesis. *Neuropathol. Appl. Neurobiol.* 28, 343–357.
- Jiang, Y.L., Yu, W.L., Zhang, J.W., Frolet, C., Di Guilmi, A.M., Zhou, C.Z., Vernet, T., and Chen, Y. (2011). Structural basis for the substrate specificity of a novel beta-N-acetylhexosaminidase StrH protein from *Streptococcus pneumoniae* R6. *J. Biol. Chem.* 286, 43004–43012.
- Khiar, N., Leal, M.P., Baati, R., Ruhlmann, C., Mioskowski, C., Schultz, P., and Fernandez, I. (2009). Tailoring carbon nanotube surfaces with glyconanorings: new bionanomaterials with specific lectin affinity. *Chem. Commun. (Camb)* 21, 4121–4123.
- Kim, E.J., Perreira, M., Thomas, C.J., and Hanover, J.A. (2006). An O-GlcNAcase-specific inhibitor and substrate engineered by the extension of the N-acetyl moiety. *J. Am. Chem. Soc.* 128, 4234–4235.
- Kornfeld, S. (1987). Trafficking of lysosomal-enzymes. *FASEB J.* 1, 462–468.
- Lang, K., and Chin, J.W. (2014). Cellular incorporation of unnatural amino acids and bioorthogonal labeling of proteins. *Chem. Rev.* 114, 4764–4806.
- Lee, Y.J., Wu, B., Raymond, J.E., Zeng, Y., Fang, X.Q., Wooley, K.L., and Liu, W.R.S. (2013). A genetically encoded acrylamide functionality. *ACS Chem. Biol.* 8, 1664–1670.
- Liu, C.C., and Schultz, P.G. (2010). Adding new chemistries to the genetic code. *Annu. Rev. Biochem.* 79, 413–444.
- Liu, Y., Chan, Y.M., Wu, J., Chen, C., Benesi, A., Hu, J., Wang, Y., and Chen, G. (2011). Chemical synthesis of a bisphosphorylated mannose-6-phosphate N-glycan and its facile monoconjugation with human carbonic anhydrase ii for in vivo fluorescence imaging. *ChemBioChem* 12, 685–690.
- Liu, Y., Marshall, J., Li, Q., Edwards, N., and Chen, G. (2013). Synthesis of novel bivalent mimetic ligands for mannose-6-phosphate receptors. *Bioorg. Med. Chem. Lett.* 23, 2328–2331.
- Luzio, J.P., Pryor, P.R., and Bright, N.A. (2007). Lysosomes: fusion and function. *Nat. Rev. Mol. Cell Biol.* 8, 622–632.
- Nguyen, D.P., Lusic, H., Neumann, H., Kapadnis, P.B., Deiters, A., and Chin, J.W. (2009). Genetic encoding and labeling of aliphatic azides and alkynes in

- recombinant proteins via a pyrrolysyl-tRNA synthetase/tRNA(CUA) pair and click chemistry. *J. Am. Chem. Soc.* **131**, 8720–8721.
- Noren, C.J., Anthony-Cahill, S.J., Griffith, M.C., and Schultz, P.G. (1989). A general method for site-specific incorporation of unnatural amino acids into proteins. *Science* **244**, 182–188.
- Ohsawa, M., Kotani, M., Tajima, Y., Tsuji, D., Ishibashi, Y., Kuroki, A., Itoh, K., Watabe, K., Sango, K., Yamanaka, S., et al. (2005). Establishment of immortalized Schwann cells from Sandhoff mice and corrective effect of recombinant human beta-hexosaminidase A on the accumulated GM2 ganglioside. *J. Hum. Genet.* **50**, 460–467.
- Pai, J., Hyun, J.Y., Jeong, J., Loh, S., Cho, E.H., Kang, Y.S., and Shin, I. (2016). Carbohydrate microarrays for screening functional glycans. *Chem. Sci.* **7**, 2084–2093.
- Park, S., and Shin, I. (2007). Carbohydrate microarrays for assaying galactosyltransferase activity. *Org. Lett.* **9**, 1675–1678.
- Park, S., Lee, M.R., and Shin, I. (2008). Chemical tools for functional studies of glycans. *Chem. Soc. Rev.* **37**, 1579–1591.
- Park, S., Kim, G.H., Park, S.H., Pai, J., Rathwell, D., Park, J.Y., Kang, Y.S., and Shin, I. (2015). Probing cell-surface carbohydrate binding proteins with dual-modal glycan-conjugated nanoparticles. *J. Am. Chem. Soc.* **137**, 5961–5968.
- Platt, F.M. (2014). Sphingolipid lysosomal storage disorders. *Nature* **510**, 68–75.
- Priyanka, P., Parsons, T.B., Miller, A., Platt, F.M., and Fairbanks, A.J. (2016). Chemoenzymatic synthesis of a phosphorylated glycoprotein. *Angew. Chem. Int. Ed.* **55**, 5058–5061.
- Presolski, S.I., Hong, V.P., and Finn, M.G. (2011). Copper-catalyzed azide-alkyne click chemistry for bioconjugation. *Curr. Protoc. Chem. Biol.* **3**, 153–162.
- Solomon, M., and Muro, S. (2017). Lysosomal enzyme replacement therapies: historical development, clinical outcomes, and future perspectives. *Adv. Drug Deliv. Rev.* **118**, 109–134.
- Tian, X., Pai, J., Baek, K.H., Ko, S.K., and Shin, I. (2011). Fluorophore-labeled, Peptide-based glycoclusters: synthesis, binding properties for lectins, and detection of carbohydrate-binding proteins in cells. *Chem. Asian J.* **6**, 2106–2112.
- Tong, P.Y., Gregory, W., and Kornfeld, S. (1989). Ligand interactions of the cation-independent mannose 6-phosphate receptor - the stoichiometry of mannose 6-phosphate binding. *J. Biol. Chem.* **264**, 7962–7969.
- Tong, P.Y., and Kornfeld, S. (1989). Ligand interactions of the cation-dependent mannose 6-phosphate receptor. *J. Biol. Chem.* **264**, 7970–7975.
- Tsuji, D., Akeboshi, H., Matsuoka, K., Yasuoka, H., Miyasaki, E., Kasahara, Y., Kawashima, I., Chiba, Y., Jigami, Y., Taki, T., et al. (2011). Highly phosphomannosylated enzyme replacement therapy for GM2 gangliosidosis. *Ann. Neurol.* **69**, 691–701.
- Unverzagt, C., and Kajihara, Y. (2013). Chemical assembly of N-glycoproteins: a refined toolbox to address a ubiquitous posttranslational modification. *Chem. Soc. Rev.* **42**, 4408–4420.
- van Kasteren, S.I., Kramer, H.B., Jensen, H.H., Campbell, S.J., Kirkpatrick, J., Oldham, N.J., Anthony, D.C., and Davis, B.G. (2007). Expanding the diversity of chemical protein modification allows post-translational mimicry. *Nature* **446**, 1105–1109.
- Wang, L., Brock, A., Herberich, B., and Schultz, P.G. (2001). Expanding the genetic code of *Escherichia coli*. *Science* **292**, 498–500.
- Wang, L.X., and Amin, M.N. (2014). Chemical and chemoenzymatic synthesis of glycoproteins for deciphering functions. *Chem. Biol.* **21**, 51–66.
- Wang, W., Hong, S., Tran, A., Jiang, H., Triano, R., Liu, Y., Chen, X., and Wu, P. (2011). Sulfated ligands for the copper(I)-catalyzed azide-alkyne cycloaddition. *Chem. Asian J.* **6**, 2796–2802.
- Wennekes, T., van den Berg, R.J.B.H.N., Boot, R.G., van der Marel, G.A., Overkleef, H.S., and Aerts, J.M.F.G. (2009). Glycosphingolipids-nature, function, and pharmacological modulation. *Angew. Chem. Int. Ed.* **48**, 8848–8869.
- Xie, Y., Ge, J., Lei, H., Peng, B., Zhang, H., Wang, D., Pan, S., Chen, G., Chen, L., Wang, Y., et al. (2016). Fluorescent probes for single-step detection and proteomic profiling of histone deacetylases. *J. Am. Chem. Soc.* **138**, 15596–15604.
- Xie, Z., and Klionsky, D.J. (2007). Autophagosome formation: core machinery and adaptations. *Nat. Cell Biol.* **9**, 1102–1109.
- Yamaguchi, T., Amin, M.N., Toonstra, C., and Wang, L.X. (2016). Chemoenzymatic synthesis and receptor binding of mannose-6-phosphate (M6P)-containing glycoprotein ligands reveal unusual structural requirements for m6p receptor recognition. *J. Am. Chem. Soc.* **138**, 12472–12485.
- Yuan, L., Lin, W.Y., Zhao, S., Gao, W.S., Chen, B., He, L.W., and Zhu, S.S. (2012). A unique approach to development of near-infrared fluorescent sensors for in vivo imaging. *J. Am. Chem. Soc.* **134**, 13510–13523.
- Zhang, G., Zheng, S.Q., Liu, H.P., and Chen, P.R. (2015). Illuminating biological processes through site-specific protein labeling. *Chem. Soc. Rev.* **44**, 3405–3417.
- Zhang, J.T., Li, C., Dutta, C., Fang, M.X., Zhang, S.W., Tiwari, A., Werner, T., Luo, F.T., and Liu, H.Y. (2017). A novel near-infrared fluorescent probe for sensitive detection of beta-galactosidase in living cells. *Anal. Chim. Acta* **968**, 97–104.
- Zhou, Q., Stefano, J.E., Harrahy, J., Finn, P., Avila, L., Kyazike, J., Wei, R., Van Patten, S.M., Gotschall, R., Zheng, X., et al. (2011). Strategies for neoglycan conjugation to human acid α -glucosidase. *Bioconjug. Chem.* **22**, 741–751.
- Zhu, Y., Li, X., Kyazike, J., Zhou, Q., Thurberg, B.L., Raben, N., Mattaliano, R.J., and Cheng, S.H. (2004). Conjugation of mannose 6-phosphate-containing oligosaccharides to acid α -glucosidase improves the clearance of glycogen in Pompe mice. *J. Biol. Chem.* **279**, 50336–50341.
- Zhu, Y., Jiang, J.L., Gumlaw, N.K., Zhang, J., Bercury, S.D., Ziegler, R.J., Lee, K., Kudo, M., Canfield, W.M., Edmunds, T., et al. (2009). Glycoengineered acid alpha-glucosidase with improved efficacy at correcting the metabolic aberrations and motor function deficits in a mouse model of Pompe disease. *Mol. Ther.* **17**, 954–963.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Mouse monoclonal anti- GM2-ganglioside	TOKYO CHEMICAL INDUSTRY CO., LTD.	Cat#A2576
Goat anti-mouse IgG-FITC	Santa Cruz Biotechnology, Inc.	Cat#SC-2010; RRID:AB_631735
Bacterial and Virus Strains		
<i>E. coli</i> DH10B	Invitrogen	Cat#C6400-03
Chemicals, Peptides, and Recombinant Proteins		
2-azidoethyl α -D-mannopyranoside	Park and Shin, 2007	N/A
Imidazole	Sigma-Aldrich	Cat#I2399; CAS: 288-32-4
tert-Butyldimethylsilyl chloride	Sigma-Aldrich	Cat#190500; CAS: 18162-48-6
Acetic anhydride	SAMCHUN	Cat#A0069; CAS: 108-24-7
Pyridine	Burdick & Jackson	Cat#UN1282; CAS: 110-86-1
Acetic acid glacial	Sigma-Aldrich	Cat#537020; CAS: 64-19-7
Tetrabutylammonium fluoride solution 1.0 M in THF	Sigma-Aldrich	Cat#216143; CAS: 429-41-4
Molecular sieves, 4 Å	Sigma-Aldrich	Cat#688363; CAS: 70955-01-0
5-Phenyl-1H-tetrazole	Sigma-Aldrich	Cat#347744; CAS: 18039-42-4
Di-t-butyl N,N-diisopropylphosphoramidite	Sigma-Aldrich	Cat#419362; CAS: 137348-86-8
Triethylamine	DAEJUNG	Cat#8556-4400; CAS: 121-44-8
Hydrogen peroxide solution	Sigma-Aldrich	Cat#95321; CAS: 7722-84-1
0.5 M Sodium methoxide in methanol	Sigma-Aldrich	Cat#403067; CAS: 124-41-4
Amberlite IR-120 hydrogen form	Sigma-Aldrich	Cat#10322; CAS: 39389-20-3
TentaGel S RAM Fmoc resin	Advanced Chem Tech	Cat#SJ5030
Fmoc-Lys(AY)	Galibert et al., 2009	N/A
N ₃ -CO ₂ H linker	Khier et al., 2009	N/A
Rh-Pip-COOH	Xie et al., 2016	N/A
Fmoc- β -alanine	TOKYO CHEMICAL INDUSTRY CO., LTD.	Cat#F0882; CAS: 35737-10-1
Fmoc-Lys(Mtt)-OH	GL Biochem	Cat#GLS180114-36813; CAS: 167393-62-6
O-(6-Chlorobenzotriazol-1-yl)-N,N,N',N'-tetramethyluronium hexafluorophosphate (HCTU)	Fluorochem	Cat#024914; CAS: 330645-87-9
N,N-Diisopropylethylamine	DAEJUNG	Cat#3078-4400; CAS: 7087-68-5
Piperidine	DAEJUNG	Cat#6646-4400; CAS: 110-89-4
Sodium L-ascorbate	Sigma-Aldrich	Cat#11140; CAS: 134-03-2
Copper(II) sulfate	Sigma-Aldrich	Cat#451657; CAS: 7758-98-7
3-[4-((Bis[(1-tert-butyl-1H-1,2,3-triazol-4-yl)methyl]amino)methyl)-1H-1, 2,3-triazol-1-yl]propanol (BTTP)	Wang et al., 2011	N/A
Trifluoroacetic acid (TFA)	Sigma-Aldrich	Cat#T6508; CAS: 76-05-1
Triethylsilane	TOKYO CHEMICAL INDUSTRY CO., LTD.	Cat#T0662; CAS: 617-86-7
Compound 5	Hyun et al., 2018	N/A
4-Chlororesorcinol	Sigma-Aldrich	Cat#C70606; CAS: 95-88-5
n-Tetrabutylammonium hydrogensulfate	Sigma-Aldrich	Cat#155837; CAS: 32503-27-8
Potassium carbonate	ACROS organics	Cat#98040010; CAS: 584-08-7
2-Azido-N-(2-morpholinoethyl)acetamide	Aurora Fine Chemicals Ltd	CAS: 1554618-67-5
Aminoguanidine hydrochloride	Sigma-Aldrich	Cat#396494; CAS: 1937-19-5

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Formaldehyde solution	Sigma-Aldrich	Cat#47608; CAS: 50-00-0
Triton™ X-100	Sigma-Aldrich	Cat#T9284; CAS: 9002-93-1
GM2 oligosaccharide	Carbosynth	Cat#OO46228
GM3 oligosaccharide	Carbosynth	Cat#OS04397; CAS: 128596-80-5
LysoTracker™ Green DND-26	ThermoFisher Scientific	Cat#L7526
D-Mannose 6-phosphate disodium salt	Sigma-Aldrich	Cat#M6876; CAS: 33068-18-7
Sodium azide	Sigma-Aldrich	Cat#S8032; Cas: 26628-22-8
O-(2-Acetamido-2-deoxy-D-glucopyranosylidenamino) N-phenylcarbamate (PUGNAc)	Carbosynth	Cat#EA06838; Cas: 132489-69-1
Lys(PA)	Nguyen et al., 2009	N/A
Sulfo-Cy5-NHS	Lumiprobe	Cat#13320
Sulfo-Cy3-NHS	Lumiprobe	Cat#11020
Human hexosaminidase B (HEXB)	Sino Biological Inc.	Cat#13169
Endoproteinase Glu-C from <i>Staphylococcus aureus</i> V8	Worthington	Cat#LS003605; CAS: 66676-43-5
Experimental Models: Cell Lines		
C2C12 myoblasts	American Type Culture Collection	Cat#ATCC® CRL-1772
NIH 3T3 fibroblasts	American Type Culture Collection	Cat#ATCC® CRL-1658
Primary Tay-Sach (TS) fibroblasts	Coriel Institute	Cat#GM00527
Primary Sandhoff (SD) fibroblasts	Coriel Institute	Cat#GM11098
Oligonucleotides		
β-N-Acetylhexosaminidase_K432TAG (forward): CTT TGG CAG TTT CGA TTA GGA CAT CAT TGT TTC TAT G	This paper	N/A
β-N-Acetylhexosaminidase_K432TAG (reverse): CAT AGA AAC AAT GAT GTC CTA ATC GAA ACT GCC AAA G	This paper	N/A
β-N-Acetylhexosaminidase_K561TAG (forward): GCA CTT AAT GAG GTG CCA TAG GAC CTG AAT AGA TAT AC	This paper	N/A
β-N-Acetylhexosaminidase_K561TAG (reverse): GTA TAT CTA TTC AGG TCC TAT GGC ACC TCA TTA AGT GC	This paper	N/A
β-N-Acetylhexosaminidase_strep-tag (reverse): C GAA TTG CGG ATG CGA CCA TCC CGA GCC GCC TGA ATT ACT CTG AAA GTA CAG ATT CTC GC	This paper	N/A
β-N-Acetylhexosaminidase_(forward): ATA TCA TGA ATG AAA AAC TGG CTA AAA AGA AGA TCG TTT CG	This paper	N/A
Recombinant DNA		
Plasmid: β-N-Acetylhexosaminidase	This paper	N/A
Software and Algorithms		
FlexAnalysis 3.4 Software	Bruker	N/A
Xcalibur™ 3.0 Software	Thermo Fisher Scientific	N/A
ZEN 2.3	Carl Zeiss	N/A

CONTACT FOR REAGENT AND RESOURCE SHARING

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Professor Injae Shin (injae@yonsei.ac.kr).

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Mouse C2C12 myoblast cells (American Type Culture Collection, Cat# ATCC® CRL-1772) and mouse NIH3T3 fibroblast cells (American Type Culture Collection, Cat# ATCC® CRL-1658) were cultured in Dulbecco modified Eagle medium (DMEM) supplemented

with 10% fetal bovine serum (FBS) at 37°C with 5% CO₂. The sex of these two cell lines is unknown. Primary human fibroblasts originated from the Tay-Sachs disease patient (Coriell Institute, Cat#GM00527, male) and primary human fibroblasts originated from the Sandhoff disease patient (Coriell Institute, Cat#GM11098, female) were cultured in Eagle's Minimum Essential Medium (EMEM) with Earle's salts and non-essential amino acids supplemented with 15% FBS at 37°C with 5% CO₂. Cells were maintained at 37°C under a humidified atmosphere containing 5% CO₂.

METHOD DETAILS

Expression and Purification of Mutant β -N-acetylhexosaminidase Containing Lys(PA)

The β -N-acetylhexosaminidase gene was amplified by PCR and then inserted between BspHI and KpnI sites of pBAD/Myc-His to generate pBAD- β -N-acetylhexosaminidase. An amber codon (TAG) was introduced at a position of 432 or 561 of β -N-acetylhexosaminidase by site-directed mutagenesis according to the manufacturer's protocol (Invitrogen, California, USA). The plasmid containing each amber mutation was co-transformed with pEvol-BuKRS into *E. coli* DH10 β . Cells were grown in lysogeny broth (LB) supplemented with ampicillin (100 μ g/mL) and chloramphenicol (35 μ g/mL). The starter culture (2 mL) was transferred to a defined medium (100 mL, 50 mM Na₂HPO₄, 50 mM KH₂PO₄, 25 mM (NH₄)₂SO₄, 2 mM MgSO₄, 0.1% trace metals, 0.5% glycerol, 0.05% glucose and 5% amino acids) supplemented with ampicillin (100 μ g/mL), chloramphenicol (35 μ g/mL) and 3 mM Lys(PA) at 20°C. Protein expression was induced by adding 0.2% L-arabinose when optical density reached 0.8, and the culture was grown overnight at 20°C. Cells were harvested by centrifugation at 10,000 rpm for 5 min at 4°C, and stored at -80°C until further processing. Cell pellets were resuspended in a lysis buffer (100 mM Tris-HCl, 150 mM NaCl and 1 mM EDTA, pH 8.0) and sonicated. Target proteins were purified by Strep-Tactin affinity chromatography under native conditions according to the manufacturer's protocol (Novagen). Purified proteins were dialyzed against a phosphate buffer (100 mM KH₂PO₄, pH 7.0) three times to remove EDTA. Concentrations of proteins (WT-Hex, Hex-K432Lys(PA) and Hex-K561Lys(PA)) were determined by measuring absorbance at 280 nm using the molar extinction coefficient ($7.323 \times 10^4 \text{ cm}^{-1} \text{ M}^{-1}$) of β -N-acetylhexosaminidase.

Analysis of Incorporation of Lys(PA) into Proteins by MS

Lys(PA) incorporated enzymes (5 mg/mL) in 50 mM phosphate buffer (pH 8.0) containing 300 mM NaCl were digested with Glu-C (20 μ M) for 2 hr at 37°C. The digested peptides were desalted by using a C-18 spin column (Bio-Rad). The digested peptides were mixed with α -cyano-4-hydroxycinnamic acid (CHCA) matrix (10 mg/mL in water containing 50% acetonitrile and 1% trifluoroacetic acid) and subjected to MS analysis (Bruker Autoflex, Bruker Daltonics). Full-length of Lys(PA) incorporated enzymes were analyzed by using the hybrid ultrahigh resolution linear ion-trap orbitrap mass spectrometer (LTQ Orbitrap XL, Thermo Fisher Scientific).

Ligation of M6P-appended N₃-peptides to Lys(PA)-Incorporated β -N-acetylhexosaminidase by Click Chemistry

To a solution of 20 μ M of Hex-K432Lys(PA) or Hex-K561Lys(PA) in 100 mM KH₂PO₄ (pH 7.0) were sequentially added N₃-(M6P)₃-Rh (1 mM), a premixed BTTP-CuSO₄ complex ([BTTP] = 3 mM, [CuSO₄] = 1 mM), freshly prepared sodium ascorbate (2 mM), and amino-guanidine (2 mM) at room temperature. After shaking for 1 hr, the ligated enzyme was isolated by using a Micro Bio-Spin column (Bio-Rad, USA) according to the manufacturer's protocol. The isolated enzyme was stored in PBS buffer (pH 7.4) at 4°C. The ligation efficiency was determined by using 8% SDS-PAGE. WT-Hex (20 μ M) was reacted with sulfo-Cy3 NHS (30 μ M) for 30 min at room temperature and the resulting Cy3-WT-Hex was used as a control.

For determination of the efficiency of click ligation, the reaction mixture obtained after click chemistry was reacted with sulfo-Cy5 NHS (30 μ M) for 30 min at room temperature. As a control, 20 μ M of Hex-K561Lys(PA) was also reacted with sulfo-Cy5 NHS (30 μ M) for 30 min at room temperature. The labeled mixture was subjected to SDS-PAGE. Fluorescence intensity of Cy5 linked to enzymes in a protein gel was analyzed by using Gel Documentation and Analysis system (Syngene).

Enzyme Kinetics

A standard 120 μ L assay was conducted at 37°C in 50 mM sodium phosphate buffer (pH 5.0) containing various concentrations (2.5–500 μ M) of GalNAc-NIR-MP. The reaction was initiated by the addition of 50 nM of WT-Hex, Hex-K432Lys(PA), Hex-K561Lys(PA) or Hex-K561Lys-(M6P)₃-Rh to the above assay solution. Changes in fluorescence intensity of an NIR dye was detected using an LS 55 fluorometer (Perkin-ElmerTM Life Sciences) (λ_{ex} = 690 nm, λ_{em} = 710 nm). Assays were performed in triplicate. By plotting the initial reaction velocity as the function of a concentration of GalNAc-NIR-MP, a typical Michaelis-Menten curve was obtained. The kinetic parameters (k_{cat} and K_{m}) were determined by non-linear fitting of this plot using the Michaelis-Menten equation.

Cleavage of GM2-oligosaccharide to GM3-oligosaccharide by β -N-Acetylhexosaminidase

WT-Hex, Hex-K561Lys(PA) and Hex-K561Lys-(M6P)₃-Rh (10 nM) were individually added to a solution of GM2 oligosaccharide (300 nM) in 50 mM sodium phosphate buffer (pH 5.0) in a final volume of 20 μ L at 37°C. The progress of enzymatic hydrolysis was monitored by thin layer chromatography (ethyl acetate: ethanol: water = 4:2:1). The reaction was terminated by addition of MeOH (20 μ L).

General Method to Quantitate Fluorescence Intensity of Fluorescent Dyes in Cells

Fluorescence intensities of fluorescent dyes in cells after treatment with M6P-appended peptides, Hex-K561Lys-(M6P)₃-Rh, fluorogenic substrates and antibody were quantified for each image using the mean region of interest (ROI) tool with the ZEN software. Specifically, a constant circular ROI was chosen to encompass the cell of interest and the same ROI area size was used for background subtraction. All confocal microscopy images were representative of at least three independent experiments.

Internalization of M6P-appended Peptides into Cells

C2C12 cells, NIH 3T3 cells, TS fibroblasts and SD fibroblasts in culture media were treated with 10 μ M of each M6P-appended peptide (N₃-(M6P)_n-Rh, n = 1, 2, 3, 6) at 37°C for indicate times. After washing twice with DPBS to remove remaining peptides, cells in DPBS were treated with 200 nM LysoTracker Green for 15 min at 37°C and then with 1 μ g/mL of Hoechst 33342 for 5 min at 37°C. After washing twice with DPBS to remove dyes, internalization of peptides into cells was detected by imaging cells with confocal fluorescence microscopy (LMS 800, Zeiss, Germany).

For the mannose-6-phosphate competition study, cells in the culture media were pre-treated with 2 mM mannose-6-phosphate at 37°C. After incubation for 1 h, cells in the culture media were treated with 10 μ M of N₃-(M6P)₃-Rh for 4 h. After washing twice with DPBS to remove remaining peptides, cells in DPBS were treated with 200 nM LysoTracker Green for 15 min, and then with 1 μ g/mL of Hoechst 33342 for 5 min. Cells were imaged by using confocal fluorescence microscopy.

For the endocytosis inhibition study, cells in the culture media were pre-treated with 0.1% w/v NaN₃ for 1 h and then incubated with 10 μ M of N₃-(M6P)₃-Rh for 4 h. After washing twice with DPBS to remove remaining peptides, cells in DPBS were treated with 200 nM LysoTracker Green for 15 min, and then treated with 1 μ g/mL of Hoechst 33342 for 5 min. Cells were imaged by using confocal fluorescence microscopy. Fluorescence intensities of rhodamine linked to peptides in cells were quantitated according to the general method described above.

Internalization of Hex-K561Lys-(M6P)₃-Rh into Cells

Primary TS and SD fibroblasts in culture media were individually treated for indicated times at 37°C with 50 nM Hex-K561Lys-(M6P)₃-Rh or 50 nM of rhodamine-labeled, commercialized human HEXB which was produced in HEK293 cells. After washing twice with DPBS to remove remaining enzyme, cells in DPBS were treated with 200 nM LysoTracker Green for 15 min, and then treated with 1 μ g/mL of Hoechst 33342 for 5 min. Cell images were obtained by using confocal fluorescence microscopy.

For the M6P competition study, cells in the culture media were pre-treated with 2 mM M6P for 1 h at 37°C and then incubated with 100 nM Hex-K561Lys-(M6P)₃-Rh for 10 h. For the endocytosis inhibition study, cells in culture media were pre-treated with NaN₃ (0.1% w/v) for 1 h at 37°C and then incubated with 100 nM Hex-K561Lys-(M6P)₃-Rh for 10 h. After washing twice with DPBS to remove remaining enzyme, cells were imaged by using confocal fluorescence microscopy. Fluorescence intensities of rhodamine linked to enzymes in cells were quantitated according to the general method described above.

Efficiency of Fluorogenic Substrates to Detect Intracellular β -N-acetylhexosaminidase Activity

C2C12 cells in culture media were treated with 20 μ M of GalNAc-NIR-MP or GalNAc-NIR-(M6P)₆-Rh at 37°C for 4 h. After washing twice with DPBS to remove remaining probe, cells in DPBS were treated with 200 nM LysoTracker Green for 15 min, and then treated with 1 μ g/mL of Hoechst 33342 for 5 min. The cells were imaged by using confocal fluorescence microscopy. Fluorescence intensities of an NIR dye cleaved from either GalNAc-NIR-MP or GalNAc-NIR-(M6P)₆-Rh in cells were quantitated according to the general method described above.

Detection of β -N-acetylhexosaminidase Activity in Cells Using GalNAc-NIR-MP

C2C12 cells, NIH 3T3 cells, TS fibroblasts and SD fibroblasts in culture media were treated with 20 μ M of GalNAc-NIR-MP for indicated times at 37°C. After washing twice with DPBS to remove remaining probe, cells in DPBS were treated with 200 nM LysoTracker Green for 15 min, and then treated with 1 μ g/mL of Hoechst 33342 for 5 min. Cells were imaged by using confocal fluorescence microscopy.

For the inhibition study, cells in culture media were pre-treated with 50 μ M PUGNAc for 12 h at 37°C and then treated with 20 μ M of GalNAc-NIR-MP for 4 h. After washing twice with DPBS to remove remaining probe, cells in DPBS were treated with 200 nM LysoTracker Green for 15 min, and then treated with 1 μ g/mL of Hoechst 33342 for 5 min. Cells were imaged by using confocal fluorescence microscopy.

For time-dependent detection of β -N-acetylhexosaminidase activity in cells using GalNAc-NIR-MP, C2C12 cells in culture media were treated with 200 nM LysoTracker Green for 15 min at 37°C and then treated with 20 μ M GalNAc-NIR-MP. Every 5 min during a 2 h period, live cell images were obtained by using confocal fluorescence microscopy at 37°C. Fluorescence intensities of an NIR dye cleaved from GalNAc-NIR-MP in cells were quantitated according to the general method described above.

Detection of Hex-K561Lys-(M6P)₃-Rh Activity Internalized into Diseased Cells Using GalNAc-NIR-MP

Primary TS and SD fibroblasts in culture media were incubated with 100 nM Hex-K561Lys-(M6P)₃-Rh for 10 h at 37°C. After washing twice with DPBS to remove remaining enzyme, cells in culture media were treated with 20 μM GalNAc-NIR-MP for 4 h. After washing twice with DPBS to remove remaining probe, cells in DPBS were treated with 200 nM LysoTracker Green for 15 min and then with 1 μg/mL of Hoechst 33342 for 5 min. The cells were imaged by using confocal fluorescence microscopy.

For the inhibition study, primary TS and SD fibroblasts in culture media were pretreated with either 50 μM PUGNAc or 10 μM hydroxychloroquine (HCQ) for 12 h at 37°C. After washing twice with DPBS to remove remaining substances, cells in culture media were incubated with 100 nM Hex-K561Lys-(M6P)₃-Rh for 10 h at 37°C. After washing twice with DPBS to remove remaining enzyme, cells in culture media were treated with 20 μM of GalNAc-NIR-MP for 4 h. After washing twice with DPBS to remove remaining probe, cells in DPBS were imaged by using confocal fluorescence microscopy. Fluorescence intensities of fluorescent dyes in cells were quantitated according to the general method described above.

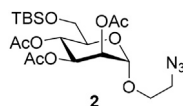
Degradation of GM2-ganglioside Accumulated in Diseased Cells by Hex-K561Lys-(M6P)₃-Rh

Primary TS and SD fibroblasts in culture media were incubated with 200 nM Hex-K561Lys-(M6P)₃-Rh for the indicated times at 37°C. After every 24 h, culture media were exchanged with fresh media containing 200 nM Hex-K561Lys-(M6P)₃-Rh. The treated cells were fixed with 4% paraformaldehyde, permeabilized with 0.25% (v/v) Triton X-100, and immunostained with mouse GM2-ganglioside antibody (1:250) followed by the FITC-labeled secondary antibody (1:1000). Cells were imaged by using confocal fluorescence microscopy. Fluorescence intensities of fluorescent dyes in cells were quantitated according to the general method described above.

General for Synthesis

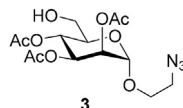
Analytical thin-layer chromatography (TLC) was conducted on silica gel 60 F254 glass plates. Compound spots were visualized by UV light (254 nm) and/or by staining with 10 wt% phosphomolybdic acid in ethanol. Flash column chromatography was performed using silica gel 60 (230–400 Mesh). NMR spectra were recorded on a Bruker DRX-400 instrument. Mass spectra were obtained by using a Bruker microflex MALDI-TOF mass spectrometer or Waters 3100 LC/MS System. UV and fluorescence spectra were recorded on JASCO V-650 and Perkin Elmer LS 55, respectively.

Synthesis



2-Azidoethyl 2,3,4-tri-O-acetyl-6-O-(tert-butyldimethylsilyl)-α-D-mannopyranoside (2)

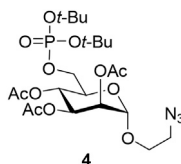
To a stirred solution of 2-azidoethyl α-D-mannopyranoside (**1**, 0.23 g, 0.96 mmol) and imidazole (0.13 g, 2 mmol) in anhydrous DMF (5 mL) was slowly added tert-butyldimethylsilyl chloride (TBSCl, 0.16 g, 1.05 mmol) with nitrogen gas flushing at room temperature. After stirring for 1 h, pyridine (10 mL) was added to the reaction mixture and acetic anhydride (0.6 g, 9.6 mmol) was then dropwise added to the mixture at 0°C. After stirring for 2 h at room temperature, the reaction was quenched by addition of saturated aqueous NaHCO₃. The aqueous layer was extracted with EtOAc three times. The combined organic layers were washed with water, dried over anhydrous Na₂SO₄, filtered and concentrated under reduced pressure. The residue was purified by flash column chromatography (hexane: EtOAc = 5:1 to 4:1) to give a product. Yield: 0.28 g (60%); ¹H NMR (400 MHz, CDCl₃): δ 5.38 (dd, 1 H, *J* = 10.7, 2.3), 5.32 (t, 1 H, *J* = 3.7 Hz), 5.27–5.25 (m, 1 H), 4.86 (d, 1 H, *J* = 1.8 Hz), 3.93–3.84 (m, 2 H), 3.73 (t, 2 H, *J* = 3.8 Hz), 3.69–3.62 (m, 1 H), 3.49–3.42 (m, 2 H), 2.14 (s, 3 H), 2.05 (s, 3 H), 2.00 (s, 3 H), 0.92 (s, 9 H), 0.08 (d, 6 H, *J* = 5.3 Hz); ¹³C NMR (100 MHz, CDCl₃): δ 170.2, 169.9, 169.4, 97.4, 71.7, 69.6, 69.2, 66.3, 50.3, 25.8, 20.8, 20.7, 20.6, 18.2, -5.3, -5.4; ESI-MS calcd for C₂₀H₃₆N₃O₉Si [*M* + *H*]⁺ 490.2, found 490.1.



2-Azidoethyl 2,3,4-tri-O-acetyl-α-D-mannopyranoside (3)

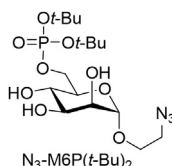
To a stirred solution of compound **2** (0.36 g, 0.74 mmol) in THF (10 mL) was added acetic acid (0.21 mL, 3.7 mmol) and *n*-Bu₄NF (1 M in THF, 1.11 mL, 1.11 mmol). After stirring for 12 h, the reaction was quenched by addition of brine (100 mL). The mixture was extracted with EtOAc three times. The combined organic layers were washed with water, dried over anhydrous Na₂SO₄, filtered and concentrated under reduced pressure. The residue was purified by flash column chromatography (hexane: EtOAc = 2:1 to 1:1) to give a product. Yield: 0.21 g (75%); ¹H NMR (400 MHz, CDCl₃): δ 5.21–5.15 (m, 2 H), 4.81 (d, 1 H, *J* = 2.0 Hz), 4.44 (dd, 1 H, *J* = 11.4, 5.3 Hz), 4.33 (dd, 1 H, *J* = 10.9, 1.4 Hz), 3.90–3.80 (m, 3 H), 3.64–3.39 (m, 2 H), 2.10 (s, 3 H), 2.09 (s, 3 H), 2.03 (s, 3 H); ¹³C NMR

(100 MHz, CDCl_3): δ 171.2, 170.7, 170.0, 97.7, 71.2, 71.0, 69.5, 66.8, 65.4, 63.3, 50.2, 24.2, 20.8, 20.7, 20.6; ESI-MS calcd for $\text{C}_{14}\text{H}_{22}\text{N}_3\text{O}_9$ [$\text{M} + \text{H}$] $^+$ 376.1, found 376.2.



2-Azidoethyl 2,3,4-tri-O-acetyl-6-O-(di-tert-butylphosphoryl)- α -D-mannopyranoside (**4**)

Compound **3** (0.37 g, 0.98 mmol), which was co-evaporated with anhydrous dioxane twice to remove residual water, was dissolved in anhydrous CH_2Cl_2 (10 mL). The mixture was added to a flame-dried flask with activated molecular sieves ($4\text{\AA}$). To a stirred mixture were added phenyltetrazole (287 mg, 2 mmol) and di-*t*-butyl *N,N*-diisopropylphosphoramidite (0.49 mL, 1.5 mmol). After stirring for 30 min, the mixture was cooled to 0°C and TEA (1 mL) and H_2O_2 (30% in water, 0.35 mL) were then added to the mixture. After stirring for 1 h at room temperature, the reaction mixture was diluted with CH_2Cl_2 . The organic layer was washed with water and saturated aqueous NaHCO_3 , dried over anhydrous Na_2SO_4 , filtered and concentrated under reduced pressure. The residue was purified by flash column chromatography (hexane: EtOAc = 1:1 to 1:2) to give a product. Yield: 0.37 g (67%); ^1H NMR (400 MHz, CDCl_3): δ 5.40 (dd, 1 H, J = 10.0, 3.6), 5.29–5.27 (m, 2 H), 4.87 (d, 1 H, J = 2.3), 4.06–4.00 (m, 3 H), 3.96–3.90 (m, 1 H), 3.67–3.62 (m, 1 H), 3.51–3.42 (m, 2 H), 2.15 (s, 3 H), 2.07 (s, 3 H), 2.01 (s, 3 H), 1.51 (s, 9 H), 1.50 (s, 9 H); ^{13}C NMR (100 MHz, CDCl_3): δ 171.2, 170.7, 170.0, 97.4, 82.8, 71.0, 69.4, 68.9, 66.9, 66.2, 65.2, 63.3, 50.2, 29.8, 20.8, 20.7, 20.6; ESI-MS calcd for $\text{C}_{22}\text{H}_{38}\text{N}_3\text{O}_{12}\text{P}$ [$\text{M} + \text{H}$] $^+$ 568.2, found 568.1.



2-Azidoethyl 6-O-(di-tert-butylphosphoryl)- α -D-mannopyranoside (**N₃-M6P(t-Bu)₂**)

To a stirred solution of compound **4** (200 mg, 0.35 mmol) in methanol (0.5 mL) was added NaOMe (0.5 M in methanol, 70 μL , 0.035 mmol). The reaction mixture was stirred for 3 h. After neutralization with Amberlite IR-120 (H^+) ion exchange resin, the reaction mixture was filtered and washed with MeOH thoroughly. The solvent was removed under reduced pressure to give a product. Yield: 0.15 g (98%); ^1H NMR (400 MHz, CDCl_3): δ 4.81 (d, 1 H, J = 2.0), 4.31–4.28 (m, 1 H), 4.09–4.06 (m, 1 H), 3.93–3.84 (m, 1 H), 3.86–3.85 (m, 1 H), 3.75 (dd, 2 H, J = 9.5, 3.5 Hz), 3.65–3.60 (m, 2 H), 3.43–3.41 (m, 2 H), 1.51 (s, 19 H); ^{13}C NMR (100 MHz, CDCl_3): δ 100.4, 83.1, 83.0, 72.0, 71.9, 70.4, 66.7, 66.6, 66.4, 50.2, 28.7; ESI-MS calcd for $\text{C}_{16}\text{H}_{33}\text{N}_3\text{O}_9\text{P}$ [$\text{M} + \text{H}$] $^+$ 442.2, found 442.3.

Solid-phase Synthesis of M6P-appended **N₃**-peptides

Peptides were synthesized by using the Fmoc/*t*Bu strategy on Tentagel Rink amide resins (0.3 mmol/g). Fmoc amino acid (3 equiv) was manually coupled on resins (5.0 mmol) in the presence of O-(6-chlorobenzotriazol-1-yl)-*N,N,N',N'*-tetramethyluronium hexafluorophosphate (HCTU, 2.8 equiv) and *N,N*-diisopropylethylamine (DIEA, 6 equiv). After each coupling reaction, the Fmoc group was removed by treatment with 20% piperidine in DMF and the resin was washed with DMF and CH_2Cl_2 several times. For click chemistry, compound **5** (2.0 equiv per one alkyne group) dissolved in DMF (400 μL) were added to peptide resins followed by addition of a solution of sodium ascorbate (1.0 equiv per one alkyne group) and CuSO_4 (1.0 equiv per one alkyne group) in water (50 μL). After shaking for 24 h, the M6P-appended peptides were cleaved from the solid support by treatment with TFA/triethylsilane/ H_2O (95:2.5:2.5, v/v/v) for 2 h. The M6P-appended peptides were purified by semi-preparative RP-HPLC (10 x 250 mm) using a gradient of 5–100% CH_3CN (0.1% TFA) in water (0.1% TFA) over 60 min, and the purified products were characterized by MALDI-TOF MS.

N₃-(M6P)₁-Rh

MALDI-MS calcd for $\text{C}_{78}\text{H}_{116}\text{N}_{18}\text{O}_{23}\text{P}$ [$\text{M} + \text{Na}$] $^+$ 1726, found 1726.

N₃-(M6P)₂-Rh

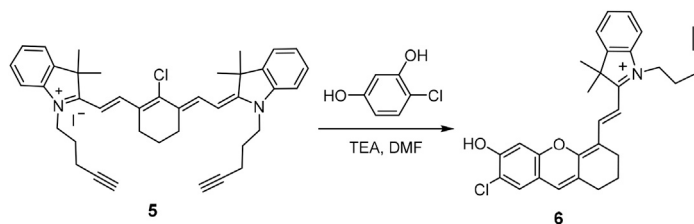
MALDI-MS calcd for $\text{C}_{100}\text{H}_{153}\text{N}_{24}\text{O}_{35}\text{P}_2$ [M] $^+$ 2312, found 2312.

N₃-(M6P)₃-Rh

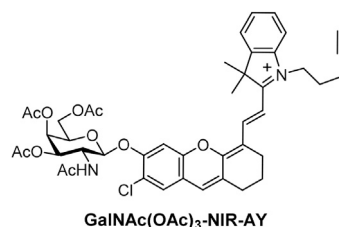
MALDI-MS calcd for $\text{C}_{122}\text{H}_{190}\text{N}_{30}\text{O}_{47}\text{P}_3$ [M] $^+$ 2920, found 2920.

***N*₃-(M6P)₆-Rh**

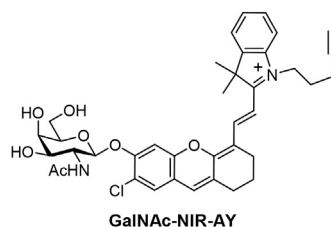
MALDI-MS calcd for C₁₈₈H₃₀₁N₄₈O₈₃P₆ [M – H][–] 4744, found 4744.

**Compound 6**

To a stirred solution of compound **5** (2.27 g, 3.17 mmol) and 4-chlororesorcinol (1.38 g, 9.52 mmol) in DMF (30 mL) was added TEA (0.81 g, 8 mmol). After stirring for 6 h at 50°C, the reaction mixture was cooled to room temperature and then diluted with ethyl acetate and washed with brine and water. The organic layer was dried over anhydrous Na₂SO₄, filtered and concentrated under reduced pressure. The residue was purified by flash column chromatography (CH₂Cl₂: MeOH = 50:1) to give a bluish green solid. Yield: 0.83 g (56%); ¹H NMR (400 MHz, CDCl₃): δ 8.09 (d, 2 H, *J* = 13.3), 7.42 (s, 1 H), 7.31–7.25 (m, 4 H), 7.08 (t, 1 H, *J* = 8.1 Hz), 6.89 (d, 1 H, *J* = 8.8), 6.68 (s, 1 H), 5.75 (d, 1 H, *J* = 13.5), 3.96 (t, 2 H, *J* = 7.0 Hz), 2.69 (t, 2 H, *J* = 5.6 Hz), 2.64 (t, 2 H, *J* = 6.4 Hz), 2.35 (td, 2 H, *J* = 6.1, 1.5 Hz), 2.13 (t, 1 H, *J* = 2.4 Hz), 1.99 (t, 2 H, *J* = 6.8 Hz), 1.91 (t, 2 H, *J* = 6.8 Hz), 1.67 (s, 6 H); ¹³C NMR (100 MHz, CDCl₃): δ 160.1, 158.0, 143.3, 139.7, 132.7, 128.2, 127.3, 122.5, 122.3, 116.4, 115.8, 107.9, 103.9, 94.4, 83.1, 70.0, 47.6, 41.7, 29.8, 28.8, 28.0, 25.4, 24.6, 21.4, 16.3; ESI-MS calcd for C₃₀H₂₉ClNO₂ [M]⁺ 470.2, found 470.1.

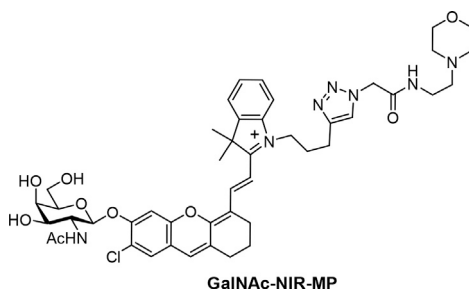
**GalNAc(OAc)₃-NIR-AY**

A solution of compound **6** (100 mg, 0.167 mmol), 2-acetamido-3,4,6-tri-O-acetyl-2-deoxy-α-D-galactopyranosyl chloride (91.7 mg, 0.25 mmol), TBAHS (57 mg, 0.167 mmol) and K₂CO₃ (46 mg, 0.334 mmol) in 1 mL of CHCl₃ and 1 mL of H₂O was vigorously stirred for 3 h at room temperature. The reaction mixture was diluted with CH₂Cl₂ and washed with H₂O. The organic layer was dried over anhydrous Na₂SO₄, filtered and concentrated under reduced pressure. The residue was purified by flash column chromatography (CH₂Cl₂: MeOH = 20:1) to afford a product as a blue solid. Yield: 42 mg (32%); ¹H NMR (400 MHz, CDCl₃): δ 8.79 (d, 1 H, *J* = 15.3), 8.75 (d, 1 H, *J* = 10.0 Hz), 7.80 (s, 1 H), 7.63 (d, 1 H, *J* = 9.1 Hz), 7.51–7.39 (m, 3 H), 7.34 (s, 1 H), 7.17 (s, 1 H), 6.52 (d, 1 H, *J* = 15.3), 6.19 (d, 1 H, *J* = 8.4 Hz), 5.63 (dd, 1 H, *J* = 11.2, 4.4 Hz), 5.55 (d, 1 H, *J* = 3.5 Hz), 4.78 (q, 1 H, *J* = 8.7 Hz), 4.52–4.45 (m, 3 H), 4.37 (dd, 1 H, *J* = 11.2, 6.4 Hz), 4.08 (dd, 1 H, *J* = 9.9, 7.4 Hz), 2.74 (t, 2 H, *J* = 6.9 Hz), 2.69 (t, 2 H, *J* = 6.4 Hz), 2.48 (td, 2 H, *J* = 6.4, 2.6 Hz), 2.20 (t, 1 H, *J* = 2.8 Hz), 2.19 (s, 3 H), 2.16 (t, 2 H, *J* = 6.9 Hz), 2.04 (s, 3 H), 2.01 (s, 3 H), 1.98 (s, 3 H), 1.91 (s, 6 H), 1.89 (s, 3 H); ¹³C NMR (100 MHz, CDCl₃): δ 178.4, 171.2, 170.4, 170.3, 170.0, 161.5, 156.2, 152.4, 146.7, 142.3, 141.0, 132.7, 129.1, 128.6, 127.6, 127.5, 123.0, 122.2, 17.1, 114.9, 112.2, 105.6, 103.6, 100.5, 82.4, 71.0, 70.7, 70.4, 66.5, 61.0, 51.2, 49.5, 44.0, 29.6, 29.1, 28.6, 28.0, 26.3, 24.2, 23.5, 20.8, 20.1, 16.0; ESI-MS calcd for C₄₄H₄₈ClN₂O₁₀ [M]⁺ 799.3, found 799.2.



GalNAc-NIR-AY

To a stirred solution of GalNAc(OAc)₃-NIR-AY (35 mg, 42 μ mol) in MeOH (500 μ L) was added NaOMe (0.5 M in methanol, 10 μ L, 4.5 mmol). The reaction mixture was stirred for 3 h at room temperature. After neutralization with Amberlite IR-120 (H⁺) ion exchange resin, the reaction mixture was filtered and washed with MeOH thoroughly. The solvent was removed under the reduced pressure to give a product. Yield: 28 mg (97%); ¹H NMR (400 MHz, CD₃OD): δ 8.83 (d, 1 H, *J* = 14.3), 7.72 (d, 1 H, *J* = 10.0 Hz), 7.76 (d, 1 H, *J* = 8.0 Hz), 7.59–7.55 (m, 2 H), 7.52 (d, 1 H, *J* = 6.7), 7.47 (s, 1 H), 7.28 (s, 1 H), 6.70 (d, 1 H, *J* = 14.3), 5.22 (d, 1 H, *J* = 8.0 Hz), 4.55 (t, 2 H, *J* = 6.7), 4.38 (t, 1 H, *J* = 9.3), 3.99–3.83 (m, 6 H), 2.78 (t, 2 H, *J* = 6.9 Hz), 2.71 (t, 2 H, *J* = 6.4 Hz), 2.58 (t, 1 H, *J* = 2.3), 2.47 (td, 2 H, *J* = 6.4, 2.6 Hz), 2.15 (t, 2 H, *J* = 6.9 Hz), 2.05 (s, 3 H), 1.87 (s, 3 H), 1.85 (s, 3 H); ¹³C NMR (100 MHz, CD₃OD): δ 178.9, 172.7, 160.4, 155.5, 152.2, 146.2, 142.5, 141.3, 129.2, 127.5, 122.6, 120.6, 117.5, 114.7, 113.1, 101.4, 82.2, 76.8, 70.4, 68.3, 61.5, 51.0, 43.9, 27.0, 26.8, 26.2, 23.7, 21.7, 15.1; ESI-MS calcd for C₂₅H₃₁N₂O₁₁ [M]⁺ 674.2, found 674.1.



GalNAc-NIR-MP

To a stirred solution of GalNAc-NIR-AY (20 mg, 28 μ mol) in DMF (0.5 mL) was added 2-azido-N-(2-morpholinoethyl)acetamide (6 mg, 28 μ mol). A solution of CuSO₄ (7 mg, 28 μ mol) and sodium ascorbate (5.5 mg, 28 μ mol) in H₂O (0.1 mL) was added to the reaction mixture. After stirring for 10 min, the mixture was directly purified by preparative reversed-phase HPLC (22 x 250 mm) using a gradient of 5–100% CH₃CN (0.1% TFA) in water (0.1% TFA) over 80 min. Yield: 11 mg (45%); ¹H NMR (400 MHz, CD₃OD): δ 8.74 (d, 1 H, *J* = 14.3), 7.92 (s, 1 H), 7.70 (d, 1 H, *J* = 6.9 Hz), 7.60–7.41 (m, 5 H), 7.23 (s, 1 H), 6.59 (d, 1 H, *J* = 14.3), 5.17 (d, 1 H, *J* = 8.0 Hz), 4.51 (t, 2 H, *J* = 6.7), 4.41 (t, 1 H, *J* = 9.3), 4.01–3.80 (m, 8 H), 3.71 (t, 2 H, *J* = 5.9 Hz), 2.98 (t, 2 H, *J* = 6.9 Hz), 2.76 (t, 2 H, *J* = 6.4 Hz), 2.67 (t, 1 H, *J* = 2.3), 2.34 (t, 2 H, *J* = 6.4 Hz), 2.02 (s, 3 H), 1.95 (t, 2 H, *J* = 6.9 Hz), 1.84 (s, 3 H), 1.80 (s, 3 H); ¹³C NMR (100 MHz, CD₃OD): δ 178.7, 172.7, 167.8, 161.5, 161.1, 160.2, 155.4, 152.2, 146.1, 145.9, 142.5, 141.2, 130.9, 129.3, 128.9, 127.5, 124.2, 122.4, 120.5, 117.5, 114.8, 112.8, 104.8, 101.4, 76.7, 71.2, 68.3, 63.5, 61.5, 56.3, 52.1, 52.0, 51.5, 51.0, 44.3, 33.5, 28.8, 27.0, 26.8, 26.5, 23.6, 21.7, 20.0; ESI-MS calcd for C₄₆H₅₇ClN₇O₉ [M]⁺ 886.4, found 886.4.

GalNAc-NIR-(M6P)₃ or 6-Rh

To a stirred solution of N₃-(M6P)₃ or 6-Rh peptide (1 μ mol) in H₂O (200 μ L) was sequentially added GalNAc-NIR-AY (0.9 mg, 2 μ mol) in DMF (0.5 mL), sodium ascorbate (100 mM in H₂O, 35 μ L, 3.5 μ mol) and CuSO₄ (100 mM in H₂O, 35 μ L, 3.5 μ mol). The resulting mixture was stirred for 1 h at room temperature. The crude product was directly purified by semi-preparative reversed-phase HPLC (10 x 250 mm) using a gradient of 5–100% CH₃CN (0.1% TFA) in water (0.1% TFA) over 60 min, and the purified products were characterized by MALDI-TOF MS.

GalNAc-NIR-(M6P)₃-Rh

MALDI-MS calcd for C₁₆₀H₂₃₂ClN₃₂O₅₄P₃ [M]⁺ 3593, found 3593.

GalNAc-NIR-(M6P)₆-Rh

MALDI-MS calcd for C₂₂₆H₃₄₃ClN₅₀O₉₀P₆ [M]⁺ 5418, found 5418.

QUANTITATION AND STATISTICAL ANALYSIS

Statistics

Data are presented as mean $\pm$ s.d. from at least three independent experiments.

DATA AND SOFTWARE AVAILABILITY

Data are available upon request to the Lead contact.